



Empirical analysis of performance assessment for imbalanced classification

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Abstract

There are multiple scenarios in machine learning where the data used presents a heavy bias towards one of the classes. Evaluating the performance of machine learning models in such imbalanced scenarios proves to be difficult and challenging, as one of the classes is poorly represented in the data, and this class is often more relevant to the end-user. An abundance of performance metrics have been devised and commonly used in order to solve these specific problems, however, there is often a lack of common agreement on which metric is best and which to use in specific imbalanced scenarios. In this study, we experimentally study the impact of choosing one metric over another in the evaluation of a classifier for binary classification, as well as the effect of data characteristics such as class imbalance and noise on those metrics. Based on our extensive empirical analysis, we provide a set of easy-to-follow guidelines for which performance metric is best to use depending on the context of the problem. Specifically, we highlight the importance of using multiple different metrics which are fundamentally different in imbalanced domains, we also display results on why the usage of Davis' interpolation of the area under the precision-recall curve and the Matthews Correlation Coefficient metrics should be preferred over other similar metrics, as well as why the usage of geometric mean and F_1 score should be avoided in scenarios likely to present noise on the labels.

Keywords Imbalanced learning · Performance metrics · Performance evaluation

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1 Introduction

Correctly evaluating the performance of any machine learning task is a difficult undertaking, as there is a vast array of metrics to choose from and the choice of a performance metric over another could lead to different outcomes in the selection of the best model. This task involves not only having a deep knowledge of the context to choose a measure representative of the desired outcome but also selecting a measure that is appropriate for the intrinsic data characteristics of the dataset. This is a well-known issue in imbalanced classification problems, where one of the classes in the data is vastly underrepresented compared to the other. Multiple application domains suffer from this problem including scenarios such as fraud detection, rare illnesses detection, intrusion detection, and many more. In these domains, there is often a particular interest toward the classification of the minority class (Branco et al., 2016). This interest needs to be correctly represented in the metric chosen and, as such, some commonly used metrics prove to not be suitable for correctly evaluating the performance of these models (Branco et al., 2016). Some metrics such as the F_β -measure, the Geometric Mean (G-Mean), or the Precision-Recall Curve (PR curve) have emerged as staples in the research community when dealing with such imbalanced problems. However, with their use slowly starting to become widespread, it raises the questions of how these metrics compare to one another and if certain of these metrics present some undesirable/desirable properties that could lead to different conclusions, depending on the specific deployment context.

We show a concrete example where the use of one performance metric over another would lead to a different outcome in Table 1, where four different classifiers.¹ were trained on an imbalanced synthetic dataset, with one of the classes representing only 10% of the total number of samples. For demonstration purposes, each classifier was evaluated using three different performance metrics, namely F_1 score, H-measure, and G-Mean, the score of which can be seen in parentheses next to the classifier name within the table. As we can observe, a researcher using the F_1 score as their metric of choice would end up choosing a different classifier than a researcher who is using G-Mean. Moreover, while the best classifier signaled by the F_1 score and H-measure is the same (Random Forest), researchers using these two metrics would end up with different classifiers in the following rank positions. As we will demonstrate in future experiments, this behavior becomes even more pronounced at higher levels of imbalance, which can become problematic when selecting a metric for evaluating an imbalanced classification problem. We aim to address this issue throughout this paper, by providing more insights that can help the researchers and end-users in the performance metrics selection process.

In our previous work (Gaudreault et al., 2021), we explored the impact of the choice of performance metrics and their implementation on the ranking of machine learning models in an imbalanced setting and concluded that the impact of choosing a different implementation of a performance metric, namely the PR curve, over another is minimal but that the usage of different types of metrics is paramount to get a proper overview of the performance of a model in an imbalanced setting. However, this preceding work mainly focused on analyzing the different ways of interpolating the precision-recall curve and how it impacts the end result when selecting a classifier based on that metric, which means it

¹ The hyperparameters used for each classifier are the defaults given by the corresponding library implementation, a summary of which is displayed in Appendix 1.

Table 1 Rankings of multiple classifiers using different performance metrics

Rank	F-1 score	H-measure	G-mean
1	Random forest (0.89)	Random forest (0.89)	Decision tree (0.93)
2	Decision tree (0.83)	XGBoost (0.87)	Random forest (0.89)
3	XGBoost (0.80)	Support vector (0.78)	XGBoost (0.86)
4	Support vector (0.33)	Decision tree (0.75)	Support vector (0.45)

was limited in scope. As such, in this work, we aim at providing more general recommendations on metric selection in imbalanced domains, by answering two main questions: (i) which performance metrics are best to use in specific scenarios; and (ii) how are the metrics typically used in an imbalanced scenario impacted by noise in the dataset. For this purpose, we extend our previous research to encompass 6 new performance metrics, namely the Class Weighted Accuracy (CWA), Kappa score, Matthews Correlation Coefficient (MCC), H-measure, Brier score loss, and log loss. Furthermore, we performed our experiments on 51 additional datasets and widen our range of experiments to include an analysis of the impact of noise on those metrics as well as new empirical results on the metrics' correlation. To the best of our knowledge, no previous work has provided a thorough, rigorous study and set of recommendations on which performance metric to use in a given specific imbalance scenario for binary classification.

Our key contributions in this publication are threefold:

- (i) Assess the impact of the choice of a performance metric on the perceived performance of a model in an imbalance context;
- (ii) Study the effect of noise combined with class imbalance on the studied performance metrics; and
- (iii) Provide detailed insights and recommendations on which performance metrics are best to use in different case studies.

The rest of this paper will be structured as follows: Sect. 2 will introduce the performance metrics that will be studied and their characteristics; Sect. 3 will present and discuss the results of the experiments determining the correlation between metrics and the impact of choosing one over another; Sect. 4 will compare the metrics' sensitivity to different types of noise and present the results; and Sect. 5 will present recommendations for which performance metrics to use in specific contexts given the previous results and reiterate the multiple conclusions reached throughout the paper.

2 Background

In this section, we review and briefly summarize the different metrics that will be used in this study. In order to describe the metrics more easily, we refer the reader to the confusion matrix represented in Table 2, which shows the prediction results of a binary classification problem, where the negative class is considered the majority class and the positive class the minority one. In the same table, we also define the recall,

Table 2 Confusion matrix of a binary classification problem

	Predicted values		
	Positive	Negative	
<i>Actual values</i>			
Positive	True positive (TP)	False negative (FN)	Recall (Sensitivity) $\frac{TP}{TP+FN}$
Negative	False positive (FP)	True negative (TN)	Specificity $\frac{TN}{TN+FP}$
	Precision $\frac{TP}{TP+FP}$	Negative predictive value $\frac{TN}{TN+FN}$	

Table 3 Characteristics of analyzed performance metrics

Class	Metric	Confusion matrix information	Type	Uses probabilities
<i>Threshold</i>				
	F_β	TP, FN, FP	Scalar	No
	G-mean	All	Scalar	No
	CWA	All	Scalar	No
	Kappa score	All	Scalar	No
	MCC	All	Scalar	No
<i>Ranking</i>				
	ROC curve	All	Graphical	Yes
	ROC AUC	All	Scalar	Yes
	H-measure	All	Scalar	Yes
	PR curve	TP, FN, FP	Graphical	Yes
	PR AUC	TP, FN, FP	Scalar	Yes
	PR-gain curve	TP, FN, FP	Graphical	Yes
	PR-gain AUC	TP, FN, FP	Scalar	Yes
<i>Probabilistic</i>				
	Brier loss	–	Scalar	Yes
	Log loss	–	Scalar	Yes

specificity, precision, and negative predictive value formulas which will ultimately be used when defining each metric.

The selected metrics are comprised of well-established metrics commonly seen as appropriate when dealing with binary classification in imbalanced domains which can be divided into three different types as described by Ferri et al. (2009): threshold, ranking, and probabilistic metrics. Each of these groups represents metrics that measure the class prediction errors, the ability to correctly separate the classes predicted, and the uncertainty of the predictions, respectively. A summary of the evaluated metrics' characteristics can be seen in Table 3 and will be described in more detail in the coming sections.

2.1 Threshold metrics

Threshold metrics are characterized by metrics that quantify the model's prediction errors. These metrics are usually represented by a single scalar value.

2.1.1 F_β measure

The F_β measure (cf. Eq. 1) (Rijsbergen, 1979) aims to represent the trade-off between precision and recall, with the value of β controlling the weight assigned to either metric. If β is 1, precision and recall have the same importance, while for values of β higher (lower) than 1, higher importance is given to recall (precision). In most cases, researchers give equal importance to both metrics ($\beta = 1$), also referred to as the F_1 -Measure or the harmonic mean of the precision and recall.

While we focused on already well-established metrics in this paper, it is worth noting that other metrics based on the F_β measure have also recently been proposed, such as the F-measure curve (Soleymani et al., 2020) which allows to better visualize a classifier performance across different parameters such as the variation in importance between precision and recall or the class skew.

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}} \quad (1)$$

2.1.2 Geometric mean

G-Mean (cf. Eq. 2) (Kubat et al., 1998) gives equal importance to both classes and is defined as the square root of the recall of each of the classes.

$$G\text{-Mean} = \sqrt{\text{sensitivity} \cdot \text{specificity}} \quad (2)$$

2.1.3 Class weighted accuracy

Likewise to G-Mean, the CWA metric (Cohen et al., 2006) also takes into account the negative class in its computation. However, it addresses one of the issues of G-Mean: the fact that both classes are weighted equally. The CWA metric, defined in Eq. 3, deals with this problem by allowing to weight the minority class differently, using a variable w ranging from [0,1].

$$CWA = w \cdot \text{sensitivity} + (1 - w) \cdot \text{specificity} \quad (3)$$

2.1.4 Kappa score

Cohen's Kappa, also referred to as the Kappa score is a measure originally designed to evaluate the agreement between multiple raters (Cohen, 1960) and can be applied to

evaluate the performance of a machine learning classifier by evaluating the agreement between the evaluated classifier and a random classifier, calculated as per Eq. 4.

$$Kappa = \frac{2(TP \cdot TN - FN \cdot FP)}{(TP + FP)(FP + TN) + (TP + FN)(FN + TN)} \quad (4)$$

2.1.5 Matthews correlation coefficient

The Matthews Correlation Coefficient (MCC) was first introduced by Matthews (1975) in a biochemistry context. It has since been increasingly used as a metric for performance evaluation in imbalanced domains. Similar to the Kappa score, it also provides a value determining the level of agreement between different raters, or in the machine learning context, between the observations and predictions.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (5)$$

2.2 Ranking metrics

While threshold metrics are trying to represent the model's prediction correctness, ranking metrics, on the other hand, will try to quantify the capability of a machine learning model to separate the classes predicted. The computation of these metrics is performed by computing scores, at different thresholds, using the probabilities produced by the model. As there are multiple scores calculated for multiple thresholds, this type of metric is usually plotted on a diagram which can help to visualize the performance of the classifier across the range of thresholds and better understand the trade-off between the measurements presented. Still, for easier comparison between multiple models, it is also common to transform the curve plotted into a scalar value, by computing the area under the curve (AUC).

2.2.1 Receiver operating characteristic curve

The Receiver Operating Characteristic Curve (ROC curve) (Egan & Egan, 1975) is one of the more well-known and most used ranking metrics. However, its use in imbalanced domains, while widespread, has been criticized by many such as Flach (2019), Davis and Goadrich (2006), Fawcett (2006), Saito and Rehmsmeier (2015) which portrayed the metric as overly optimistic in certain scenarios, especially imbalanced ones.

The ROC curve plots the score given by the recall against the False Positive Rate (FPR) as defined in Eq. 6. In order to compute its AUC, one could simply use an approximation technique such as the trapezoidal method, or the equivalent Mann–Whitney *U*-statistic (Davis & Goadrich, 2006; Mason & Graham, 2002) which are methods commonly used to do so.

$$FPR = \frac{FP}{FP + TN} \quad (6)$$

2.2.2 H-measure

Hand (2009) presents an alternative metric to the ROC AUC named the H-Measure, which aims to alleviate some of the weaknesses of the ROC curve, for instance, that it uses different misclassification cost distributions to evaluate distinct classifiers which fails to consider the context or that interpreting the AUC of intersecting ROC curves can be misleading. This new metric acknowledges that there are different costs to specific misclassifications and introduces costs for them. The estimation of the H-measure uses the same underlying recall and FPR values used to compute the ROC curve but applies a uniform cost-distribution which is the same regardless of the evaluated classifier, therefore solving one of the major aforementioned weaknesses of the ROC AUC. This distribution was initially aimed to be selected subjectively by the researchers depending on the context of their problem and reported alongside a universal distribution, however, a standard distribution for unbalanced problems has since been proposed by the authors which we will be using in this study (Hand & Anagnostopoulos, 2014).

2.2.3 Precision-recall curve

While the ROC curve uses recall against the FPR for its calculation, the PR curve instead plots the precision against the recall. Since the precision focuses more on the minority class, by using the true and false positives compared to the FPR which uses false positives and true negatives, the usage of the PR curve is usually preferred over the ROC curve when dealing with imbalanced domains. In contrast with the ROC curve, computing the AUC of the PR curve is not as straightforward. Indeed, as Davis and Goadrich (2006) describe, the fact that precision does not vary linearly with the recall means that the use of linear interpolation techniques such as the trapezoidal method would not lead to an accurate representation of the AUC and would provide over-optimistic results in some scenarios. Consequently, other solutions have been put forward in the literature to address this problem. The following sub-sections present some of these solutions.

2.2.3.1 Interpolated precision Manning et al. (2008) present a simple solution to the aforementioned problem. By simply estimating the value for the precision of any recall point r to the highest precision value of any recall point greater than r , we can eliminate the drastic change in precision from one recall point to another, and therefore allow the linear interpolation of the curve.

2.2.3.2 Average precision Defined as the sum of precision of each point weighted by the difference in recall of the previous point (Su et al., 2015), the average precision (AP) can be computed as seen in Eq. 7.

$$AP = \sum_n (recall_n - recall_{n-1}) precision_n \quad (7)$$

where n represents the n th threshold computed from the prediction scores.

2.2.3.3 Davis' interpolation Davis and Goadrich (2006) presented another alternative to computing the AUC of the PR curve, by interpolating values for the True Positive (TP) and False Positive (FP) values between two consecutive points.

Let us consider two successive points, A and B , and their corresponding underlying TP and FP values denoted as TP_A , TP_B , FP_A , and FP_B . We can calculate the number of negative values equalling to one positive value, which is denoted as the local skew (cf. Eq. 8).

$$Local\ Skew = \frac{FP_B - FP_A}{TP_B - TP_A} \quad (8)$$

Afterward, new points are created at regular intervals between A and B such that $TP = TP_A + x$ where x represents all integers between 1 and $TP_B - TP_A$. The values representing the precision and recall of these newly created points using the local skew previously calculated are represented in Eq. 9.

$$\left(\frac{TP_A + x}{TP_A + x + FP_A}, \frac{TP_A + x}{TP_A + x + FP_A + \frac{FP_B - FP_A}{TP_B - TP_A} x} \right) \quad (9)$$

Including these new points alleviates the overestimation of using linear interpolation on the PR curve when two points are far away, as such, using a linear interpolation such as the previously mentioned trapezoidal method can now be performed to estimate the PR AUC.

2.2.4 Precision-recall-gain curve

Flach and Kull (2015) presented some drawbacks of using the PR curve, such as incoherent scale assumptions and the lack of desirable properties which are present within the ROC curve such as a universal baseline and linear interpolation. To solve some of these issues, they proposed a new approach named the Precision-Recall-Gain Curve (PRG curve), which plots the PR curve in a different coordinate system as it can be seen with the equations for the precision gain (cf. Eq. 10) and recall gain (cf. Eq. 11). The new baseline is represented by π , which represents the always-positive classifier.

$$precG = \frac{precision - \pi}{(1 - \pi)precision} \quad (10)$$

$$recG = \frac{recall - \pi}{(1 - \pi)recall} \quad (11)$$

2.3 Probabilistic metrics

Rather than evaluating the correctness of class predictions, probabilistic metrics focus on evaluating the divergence from the true probabilities and are therefore more useful in cases where we would want to evaluate the unreliability of the predictions of a model and punish confident probabilities with wrong results. While the appropriateness of using such metrics in imbalanced problems has been questioned before (e.g. Japkowicz, 2013), we wanted, nevertheless, to include a few metrics of this type as a point of comparison and for completeness purposes.

2.3.1 Brier score loss

Described as the mean squared error of the predicted and true probabilities of the positive class, the Brier score loss is computed as seen in Eq. 12.

$$\text{Brier Score Loss} = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}} (y_i - p_i)^2 \quad (12)$$

where n_{samples} represents the number of samples, y_i the real value expected and p_i the predicted probability for a given sample i .

2.3.2 Log loss

Log loss, often referred to as cross-entropy loss, aims to represent the average difference between the expected and predicted probabilities.

$$\text{Log Loss} = \frac{-1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}} (y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)) \quad (13)$$

where n_{samples} represents the number of samples, y_i the real value expected and p_i the predicted probability for a given sample i .

2.4 Other scenarios

While we focus this study on binary imbalanced classification, we also wanted to provide an overview of other scenarios, namely the imbalanced multi-class and data stream cases, which can often occur in combination with an imbalance setting. These scenarios offer their own sets of challenges which we will present in the coming sections and plan to further investigate empirically in future research.

2.4.1 Multi-class

Multi-class classification, as its name implies, represents problems that have more than two classes to classify instances under. In the imbalance case, there may be multiple classes that can be categorized as a majority or minority class with various levels of data distribution skew between themselves (Tanha et al., 2020). Hence, it can be even harder to properly evaluate the performance of a classifier as the user's desired importance of correctly classifying each class is not necessarily equal for all classes. Moreover, as we will demonstrate in the coming sections, there exist multiple ways of extending the aforementioned performance metrics to the multi-class scenario, which means that a selection of a proper metric for performance evaluation becomes even more challenging than in the binary case.

2.4.1.1 F_β measure There are two main ways of extending the F_β measure to the multi-class scenario. These extensions are based on the principle of averaging the score of each class in a slightly different way, namely micro-average, and macro-average. The former, micro-

average, globally computes the total number of TP, FN, and FP over all classes (Tanha et al., 2020) to compute the precision and recall as shown in Eqs. 14 and 15 for K classes, and then compute the resulting F_β measure as we demonstrated before in Eq. 1.

$$\text{Micro-Average Precision} = \sum_{i=1}^K \frac{TP_i}{TP_i + FP_i} \quad (14)$$

$$\text{Micro-Average Recall} = \sum_{i=1}^K \frac{TP_i}{TP_i + FN_i} \quad (15)$$

Macro-average, on the other hand, gives equal importance to all classes regardless of their size, as it simply computes the unweighted mean of the F_β score of each class in a one-vs-rest fashion, as shown in Eq. 16. We must highlight that when considering the micro-averaging extension an equal weight is given to each sample while when considering the macro-average extension an equal weight is assigned to all classes regardless of their size.

$$\text{Macro-Average } F_\beta = \frac{\sum_{i=1}^K F_{\beta_i}}{K} \quad (16)$$

Another implementation of the macro-average, often described as the weighted average, uses the weight of each class in order to compute the weighted mean of the F_β measure of each class rather than the unweighted mean.

In imbalanced domains, the preferred method will depend on the context, as the macro-average will give minority classes as much importance as the majority ones, which can be desirable or not, depending on the situation.

2.4.1.2 G-mean and CWA Similar to the binary case, G-Mean can be defined for multi-class scenarios as the K th root of the product of sensitivity (recall) for each class, when considering K classes, as shown in Eq. 17.

$$G\text{-Mean} = \sqrt[K]{\prod_{i=1}^K \text{sensitivity}_i} \quad (17)$$

It is worth noting that as with the binary case, G-Mean gives equal importance to all classes regardless of their data distribution. Thus, as we did for the binary scenario, we can define the CWA metric for the multi-class scenario by adding a specific weight w_i to each class' specificity, allowing us to weigh each class differently, as shown in Eq. 18.

$$CWA = \sum_{i=1}^K w_i \cdot \text{sensitivity}_i \quad (18)$$

where $\sum_{i=1}^K w_i = 1$.

2.4.1.3 Kappa score, MCC As described by Grandini et al. (2020), MCC and Kappa score can be defined in terms of a confusion matrix C for K classes with the following subterms (Gorodkin, 2004):

- $c = \sum_k C_{kk}$ the total number of elements correctly predicted

- $s = \sum_i^K \sum_j^K C_{ij}$ the total number of elements
- $p_k = \sum_i^K C_{ki}$ the number of time that class k was predicted
- $t_k = \sum_i^K C_{ik}$ the number of times that class k truly occurred

Using these variables, we can define MCC and the Kappa score for the multi-class case as displayed in Eqs. 19 and 20, respectively.

$$MCC = \frac{c \cdot s - \sum_k^K p_k \cdot t_k}{\sqrt{(s^2 - \sum_k^K p_k^2) \cdot (s^2 - \sum_k^K t_k^2)}} \quad (19)$$

$$Kappa = \frac{c \cdot s - \sum_k^K p_k \cdot t_k}{s^2 - \sum_k^K p_k \cdot t_k} \quad (20)$$

An interesting observation and added challenge in the multi-class setting, is that MCC, which ranged from $[-1, 1]$ in the binary case, no longer has this constant minimal value and now has a dynamic minimum value which will be in the range of -1 to 0 , depending on the distribution of the true labels (Grandini et al., 2020).

2.4.1.4 ROC, H-measure, PR curve, PRG curve For ranking metrics, as they are typically only defined for binary class problems, an often accepted generalisation (Landgrebe & Duin, 2007; Hand & Till, 2001) for multi-class scenarios is to binarize the output by either using a one-vs-rest or one-vs-all scheme to compute the metric as if it was a binary problem and then using micro or macro averaging to combine them, as we previously demonstrated in the case of the F_β Measure.

2.4.1.5 Brier loss, log loss Both the Brier loss and log loss can easily be generalized to the multi-class scenario by simply summing the terms over all classes, as displayed in Eqs. 21 and 22 when considering K classes.

$$Brier\ Score\ Loss = \frac{1}{n_{samples}} \sum_{i=0}^{n_{samples}} \sum_{j=1}^K (y_{ij} - p_{ij})^2 \quad (21)$$

$$Log\ Loss = \frac{-1}{n_{samples}} \sum_{i=0}^{n_{samples}} \sum_{j=1}^K (y_{ij} \cdot \log(p_{ij}) + (1 - y_{ij}) \cdot \log(1 - p_{ij})) \quad (22)$$

2.4.2 Data streams

The imbalanced data streams domain relates to a specific type of scenario where the data arrives at the decision model in a continuous, dynamic fashion rather than as a one-time static evaluation. This field has recently gained a lot of interest as multiple real-life applications can be related to this concept, such as sensor networks, financial data, and network traffic just to name a few. In addition to the traditional difficulties of evaluating an imbalanced problem, such as the under-representation of one or more of the class(es), data streams also present other, new challenges, as the models typically evolve over time rather

than being fixed and since the data is continuous and does not come from a stationary distribution (Gama et al., 2013). A typical way of dealing with those issues is to evaluate the decision models continuously, over a sliding window of a specific number of samples using the same performance metrics we previously defined for the binary and multi-class cases (Aguiar et al., 2022; Gama et al., 2013).

Another interesting sub-domain of data streams is the problem of novelty detection. For this task, the objective of the model is to detect arriving samples that belong to a different concept than the already known ones (Faria et al., 2016). While regular performance metrics can be used to evaluate these models, the fact that correctly identifying the novelties is often more important for the end user than correctly labeling the known classes has led to the creation of domain-specific metrics such as the percentage of novel class samples misclassified as normal (M_{new}) and the percentage of normal samples misclassified as novelties (F_{new}) (Faria et al., 2016). There is, however, still a need for further research and the development of performance metrics that properly assess the performance of models in this field.

2.5 Summary

Considering the different metrics shown in this section, we will be comparing all five thresholds (F_1 -Measure, G-Mean, CWA, kappa, and MCC) and the two probabilistic metrics presented (Brier Loss, Log Loss). For the four presented ranking metrics, as one can not easily compare the curves given by most of these metrics, we will compute the AUC where it is applicable. Given that, as presented, the PR AUC can be interpolated in a myriad of ways, we will also be comparing the three proposed interpolations of the PR AUC as well as the flawed linear interpolation of the curve which gives us a total of four metrics representing the PR AUC.

3 Evaluation of the impact on rankings

In this section, we will first study the impact of using different performance metrics on the final interpretation of the performance of a model. In other words, we want to analyze how the different performance metrics used in imbalanced domains relate to each other and how the usage of one specific metric could portray a different result when compared to another one. This section is organized into four subsections that will cover: (i) a description of the methodology used when performing the experiments; (ii) an exploration of the correlation between the metrics using well-known correlation coefficients; (iii) an analysis of the choice of performance metric on the ranking of a model; and (iv) a summary of the results within this section and main takeaways.

3.1 Methodology

Based on our previous work (Gaudreault et al., 2021), we extended our experiments to be performed on a total of 66 binary datasets taken from the KEEL repository (Alcalá-Fdez et al., 2011) which are made up of a total of 20 unique datasets that are either initially binary or that were multi-class and then binarized. We must highlight that this is a common repository used by the research community for evaluating solutions for the class imbalance problem. Moreover, the binarized versions generated from a base dataset have

different characteristics, including the number of features, total number of samples, and imbalance ratio². Still, the fact that these 66 datasets are derived from an initial set of 20 might bring some limitations in terms of the generalizability of our results. These datasets were selected depending on their Imbalance Ratio (IR) (cf. Eq. 23), and were split equally into three categories: slightly ($IR < 9$), moderately ($9 \leq IR < 13$), and heavily ($IR \geq 13$) imbalanced. Table 4 displays the main characteristics of the selected datasets.

$$IR = \frac{\text{majority class samples}}{\text{minority class samples}} \quad (23)$$

The following experiments were conducted using four widely used types of machine learning models that exhibit different characteristics, namely decision tree, random forest, XGBoost, and Support Vector Machine (SVM).³ These specific classifiers were chosen because they are generally well understood and widely available in different programming language libraries, which is important for both interpretability and reproducibility purposes, respectively. In addition, we wanted to include a base classifier (decision tree), as well of variants of it which are used with ensemble methods based on both boosting (XGBoost) and bagging (random forest).

These models were evaluated using the 14 performance metrics and AUC interpolation techniques presented in Sect. 2, which represent a broad range of evaluation metrics often used in imbalanced domains, including lesser-used ones included for comparison purposes.

As we will discuss further in the coming sections, two main experiments were performed with the following three goals in mind: (i) to extend our previous work to include new metrics and datasets; (ii) to establish if a correlation exists between the different performance metrics evaluated; and (iii) to analyze the impact of the ranking given by a specific metric compared to another. These experiments will help us garner more information on the strengths and weaknesses of each metric, in order to provide more general recommendations accordingly. The complete results, as well as the code to reproduce these experiments, are freely available at <https://github.com/jgaud/PerformanceEvaluationImbalancedClassification>.

3.2 Correlation between metrics

In order to evaluate the correlation of the metrics under different conditions, we used an approach similar to the one used by Ferri et al. (2009). We computed both the Pearson and Spearman correlation matrices on each dataset independently of one another, the former evaluates the linear correlation between two variables (Pearson, 1896) while the latter evaluates how well two variables can be represented with a monotonic relation using the ranked values rather than the absolute values (Spearman, 1987). Since each dataset can display widely different characteristics such as a different number of samples, imbalance ratio, class separation, etc. which in return influence the value of the

² For instance, the “ecoli1” and the “ecoli-0-1-4-6 vs 5” are both obtained from the base dataset ecoli, but have different characteristics: the “ecoli1” has 7 features, a total of 336 samples, and an imbalance ratio of 3.36, while the “ecoli-0-1-4-6 vs 5” has 6 features, 280 samples, and an imbalance ratio of 13. Other examples, such as the different versions of the glass dataset also present a varying total number of samples (ranging from 92 to 214) and imbalance ratios (ranging from 1.82 to 22.78).

³ The hyperparameters used for each classifier are the defaults given by the corresponding library implementation, a summary of which is displayed in Appendix 1.

Table 4 Main characteristics of the 66 selected datasets ordered by increasing IR

Name	Nr. Features	Nr. cases	IR
glass1	9	214	1.82
wisconsin	9	683	1.86
ecoli-0_vs_1	7	220	1.86
pima	8	768	1.87
iris0	4	150	2
glass0	9	214	2.06
yeast1	8	1484	2.46
haberman	3	306	2.78
vehicle2	18	846	2.88
vehicle1	18	846	2.9
glass-0-1-2-3_vs_4-5-6	9	214	3.2
vehicle0	18	846	3.25
vehicle3	18	846	3.25
ecoli1	7	336	3.36
newthyroid2	5	215	5.14
new-thyroid1	5	215	5.14
ecoli2	7	336	5.46
segment0	19	2308	6.02
glass6	9	214	6.38
yeast3	8	1484	8.1
ecoli3	7	336	8.6
page-blocks0	10	5472	8.79
ecoli-0-3-4_vs_5	7	200	9
yeast-2_vs_4	8	514	9.08
ecoli-0-6-7_vs_3-5	7	222	9.09
ecoli-0-2-3-4_vs_5	7	202	9.1
glass-0-1-5_vs_2	9	172	9.12
yeast-0-3-5-9_vs_7-8	8	506	9.12
yeast-0-2-5-7-9_vs_3-6-8	8	1004	9.14
yeast-0-2-5-6_vs_3-7-8-9	8	1004	9.14
ecoli-0-4-6_vs_5	6	203	9.15
ecoli-0-1_vs_2-3-5	7	244	9.17
ecoli-0-2-6-7_vs_3-5	7	224	9.18
glass-0-4_vs_5	9	92	9.22
ecoli-0-3-4-6_vs_5	7	205	9.25
ecoli-0-3-4-7_vs_5-6	7	257	9.28
yeast-0-5-6-7-9_vs_4	8	528	9.35
vowel0	13	988	9.98
ecoli-0-6-7_vs_5	6	220	10
glass-0-1-6_vs_2	9	192	10.29
led7digit-0-2-4-5-6-7-8-9_vs_1	7	443	10.97
glass-0-1-4-6_vs_2	9	205	11.06
glass2	9	214	11.59
cleveland-0_vs_4	13	177	12.62
ecoli-0-1-4-6_vs_5	6	280	13
dermatology-6	34	358	16.9

Table 4 (continued)

Name	Nr. Features	Nr. cases	IR
zoo-3	16	101	19.2
shuttle-6_vs_2-3	9	230	22
glass5	9	214	22.78
winequality-red-4	11	1599	29.17
poker-9_vs_7	10	244	29.5
yeast-1-2-8-9_vs_7	8	947	30.57
abalone-3_vs_11	8	502	32.47
winequality-white-9_vs_4	11	168	32.6
winequality-red-8_vs_6	11	656	35.44
abalone-17_vs_7-8-9-10	8	2338	39.31
winequality-white-3_vs_7	11	900	44
winequality-red-8_vs_6-7	11	855	46.5
abalone-19_vs_10-11-12-13	8	1622	49.69
winequality-white-3-9_vs_5	11	1482	58.28
poker-8-9_vs_6	10	1485	58.4
shuttle-2_vs_5	9	3316	66.67
winequality-red-3_vs_5	11	691	68.1
abalone-20_vs_8-9-10	8	1916	72.69
poker-8-9_vs_5	10	2075	82
poker-8_vs_6	10	1477	85.88

performance metrics evaluated, it is important to evaluate the correlation matrix of each dataset independently. Subsequently, we compute the average of these correlations over all datasets, and over each imbalance group (slightly, moderately, and highly imbalanced). Since both methods (Pearson and Spearman) led to similar conclusions and for ease of reading, we only included the Spearman rank correlation matrix over all datasets, as shown in Fig. 1. The remaining correlation matrices depicting specific imbalance groups, as well as the ones created using the Pearson correlation can be observed in Appendix 2, which includes Figs. 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 and 26.

Moreover, to display the results of the coefficient matrices in a way where the relations between the different clusters can easily be seen and interpreted, we performed hierarchical clustering using the average method, as described in Müllner (2011). This method specifically, was chosen as we found its results interesting and because this choice was also supported by another analysis of metrics which was performed previously (Ferri et al., 2009). The results of these clusterings are shown in dendrograms corresponding to each of the aforementioned coefficient matrices and can be seen in Figs. 2, 3, 4 and 5, which correspond to the overall results and each group results starting from the less imbalanced to the most imbalanced datasets.

3.2.1 Discussion of results

A first conclusion that can be drawn from these results is that there generally exist three main clusters of metrics that show up in every imbalance group and which display a high correlation between themselves. Each of these clusters corresponds to a specific group of

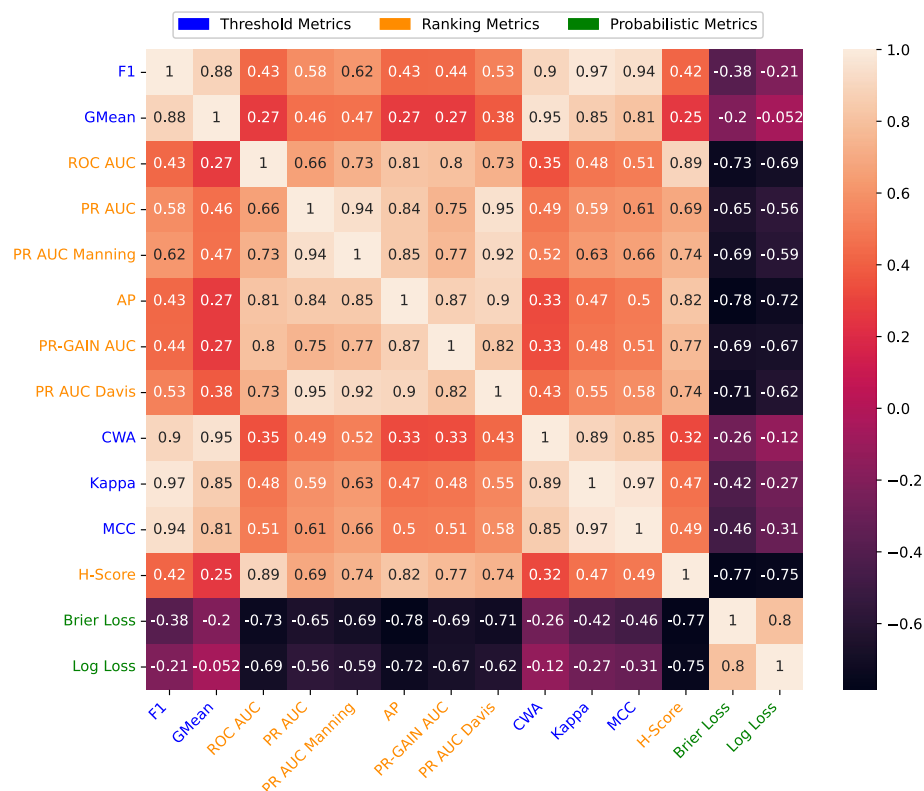


Fig. 1 Spearman correlation coefficients of all evaluated metrics over all datasets

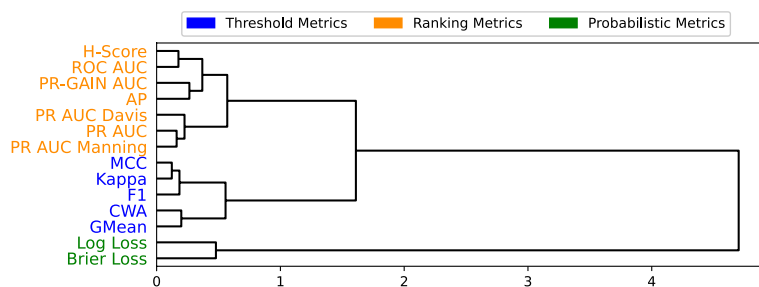


Fig. 2 Hierarchical clustering of all evaluated metrics over all datasets

metrics presented in Sect. 2, that is threshold, ranking, and probabilistic metrics. In effect, this first conclusion is in agreement with our expectations and past conclusions (Gaudreault et al., 2021). These different groups of metrics exhibit different properties due to their very nature and thus they evaluate the performance of a model from a different perspective when we compare the three groups.

Looking more closely at the correlations between specific metrics, the different PR curves implementations unsurprisingly display a high correlation between themselves. We also

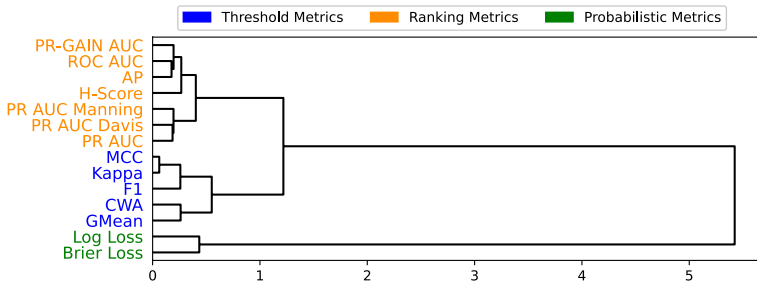


Fig. 3 Hierarchical clustering of all evaluated metrics over datasets with $IR < 9$

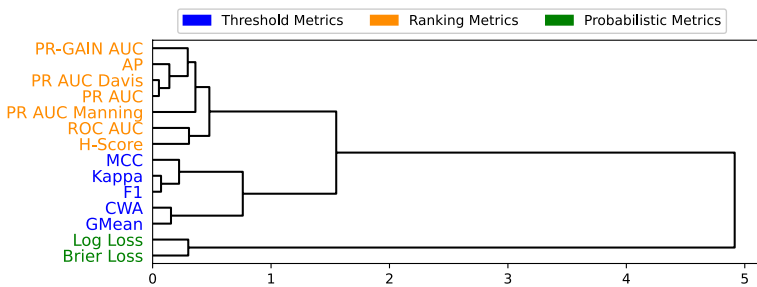


Fig. 4 Hierarchical clustering of all evaluated metrics over datasets with $9 \leq IR < 13$

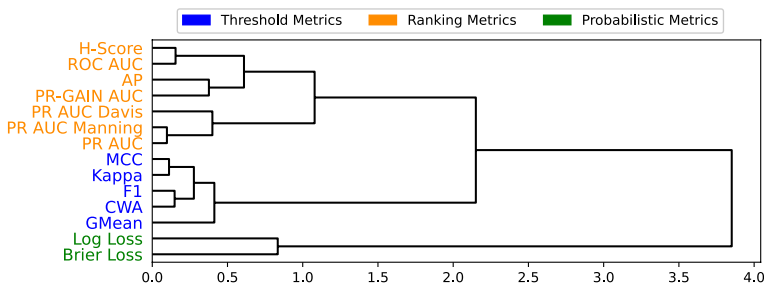


Fig. 5 Hierarchical clustering of all evaluated metrics over datasets with $IR \geq 13$

notice the existence of a separate group within the threshold metrics when looking at the correlations between MCC, Kappa, and F_1 . While these three metrics are highly correlated, we observe more specifically that the first two metrics present a high level of correlation between themselves across all groups of imbalance, even at the highest ratio. Investigating further, we can look at the pairwise comparison between the two metrics as shown in Fig. 6 which display a clear linear relationship between the two metrics, although Kappa seems to give a lower score than MCC in some cases. This is a really interesting observation, as both metrics effectively measure correlation or agreement between classifiers. It has been shown (Delgado & Tibau, 2019; Chicco et al., 2021), however, that while these two metrics can give similar results in most scenarios, Kappa can exhibit undesirable behavior in imbalance scenarios and

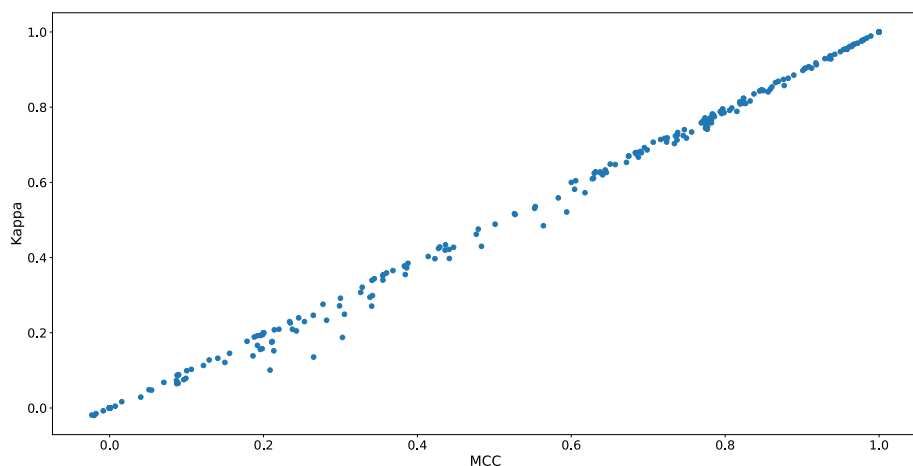


Fig. 6 Pairwise comparison between MCC and Kappa

should therefore be avoided. Indeed, in some cases which occur commonly with imbalanced data, such as when true positives and true negatives are zero, the Kappa score can provide a higher score for worse classification results and therefore lead to wrong conclusions.

Inspecting the results from the hierarchical clustering (cf. Fig. 2), we can observe from the hierarchical clustering that both the threshold and ranking metrics are at a much closer linkage distance than the two evaluated probabilistic metrics (Log Loss and Brier Loss). Indeed, the fact that the latter group of metrics uses the model's probabilities rather than the number of correct/incorrectly predicted labels like the first two groups would likely explain the higher divergence.

Another interesting observation related to the linkage length of the different dendrograms representing each dataset imbalance group (cf. Figs. 3, 4 and 5) is that the distance between the different clusters within the threshold and ranking metrics tends to increase with the increase in imbalance, while the previously mentioned distance between probabilistic metrics and the two first groups decreases. This behavior shows that clusters that are highly correlated, such as MCC and Kappa, still are at higher imbalance ratios, while other clusters that are less correlated, exhibit a higher variance at higher degrees of imbalance. This reiterates the importance of using multiple metrics, especially in highly imbalanced scenarios, as we can see that these metrics might drift apart and offer different interpretations, a conclusion we will explore further in the following section.

3.3 Impact of the choice of performance metrics

In this section, we repeat a part of the experiments performed in our previous work with a more extensive set of performance metrics and datasets. This will allow us to observe the validity of our previous results due to the use of more datasets, but also extend our insights on how the newly added metrics behave in comparison to the previous ones. The main objective of these experiments is to evaluate if the use of one performance metric over another provides a significant disparity in the ranking of multiple machine learning models at different imbalance levels. Therefore, the results will be presented in the form

of rankings rather than the absolute values of the metrics. For a given metric, each one of the k evaluated models will be given a rank from 1 to k depending on the score given by the metric. In our experiments we considered 4 machine learning algorithms (decision tree, random forest, XGBoost, and SVM), thus we will have ranks varying from 1 to 4.

3.3.1 Assessing the impact of the performance metrics

While we can evaluate the performance of classifiers using performance metrics, evaluating the metrics themselves is a difficult task, as there are few ways to objectively measure if a metric is behaving as expected. As such, in order to assess the impact of using a specific performance metric rather than another, we used two benchmark metrics, namely the Krippendorff's Alpha score and the Mode Difference (MD), as described in the following sections. These will allow us to measure the agreement between the different metrics and which metric deviates from the rankings given by others. It is worth noting, however, that a higher or lower level agreement between a specific metric compared to others does not necessarily mean that the metric is good or bad, but rather that it gives a similar, or different viewpoint which can be desirable depending on the context.

3.3.1.1 Mode difference As defined in Gaudreault et al. (2021), MD (cf. Eq. 25) is a measure used to efficiently observe how the ranking given by an observer, in our case a performance metric, differs from the general consensus of a set of observers. Assuming n performance metrics evaluating k classifiers, the rankings for each metric $m \in \{1, \dots, n\}$ on each classifier is denoted by $r_{m,1}, r_{m,2}, \dots, r_{m,k}$. This allows us to calculate the mode of the rankings for each classifier (cf. Eq. 24), in other words, the most frequent rank obtained for a classifier, or the average of the most frequent values when there are ties.

$$Mo_j = \text{Avg}(\text{Mode}(r_{1,j}, r_{2,j}, \dots, r_{n,j})) \quad (24)$$

$$MD_{i,j} = \|r_{i,j} - Mo_j\| \quad (25)$$

Finally, considering k classifiers, we compute the average of mode differences over a performance metric i (cf. Eq. 26) in order to give an average value of the variability of the rankings of the metric over all the tested classifiers. In other words, the higher this value is, the more a given metric differs from the consensus of the other tested metrics.

$$MD_AVG_i = \text{Avg}(MD_{i,1}, MD_{i,2}, \dots, MD_{i,k}) \quad (26)$$

The value of MD_AVG_i for a performance metric i varies between 0 and $k - 1$ where the lower bound signifies that the given metric agrees all the time with the consensus of the other metrics and the upper bound means that it disagrees the most.

3.3.1.2 Krippendorff's Alpha As defined by Krippendorff (2011), the Krippendorff's alpha is a metric that measures the level of agreement between multiple raters, or in our case, performance metrics. The value ranges from 0 to 1, depending on whether all raters disagree or agree, respectively.

Table 5 Ranks and mode difference of each performance metrics on the Wisconsin dataset (best performing model for each metric is underlined)

Metric	Decision tree		Random forest		XGBoost		SVM		MD_{AVG}
	Rank	MD	Rank	MD	Rank	MD	Rank	MD	
F_1	4	0	2	0	3	1	<u>1</u>	2	0.75
G-mean	4	0	2	0	3	1	<u>1</u>	2	0.75
ROC AUC	4	0	2	0	<u>1</u>	1	3	0	0.25
PR AUC (linear interpolation)	4	0	2	0	<u>1</u>	1	3	0	0.25
Interpolated precision AUC	4	0	2	0	<u>1</u>	1	3	0	0.25
AP	4	0	2	0	<u>1</u>	1	3	0	0.25
PR-gain AUC	4	0	2	0	<u>1</u>	1	3	0	0.25
PR AUC Davis	4	0	2	0	<u>1</u>	1	3	0	0.25
CWA	4	0	2	0	3	1	<u>1</u>	2	0.75
Kappa	4	0	2	0	3	1	<u>1</u>	2	0.75
MCC	4	0	2	0	3	1	<u>1</u>	2	0.75
H-measure	4	0	<u>1</u>	1	3	1	2	1	0.75
Brier loss	4	0	2	0	3	1	<u>1</u>	2	0.75
Log loss	4	0	2	0	<u>1</u>	1	3	0	0.25

Table 6 MD of all metrics averaged by imbalance class (best score by imbalance level is underlined; best overall score is in boldface)

Metric	IR < 9	9 <= IR < 13	13 <= IR
F_1	0.4261	0.5284	0.625
G-mean	0.6534	0.7556	0.7159
ROC AUC	0.267	0.3579	0.5227
PR AUC (linear interpolation)	0.2329	0.267	0.3068
Interpolated precision AUC	0.1647	<u>0.1988</u>	0.25
AP	0.2556	0.2897	0.375
PR-gain AUC	0.267	0.3806	0.4431
PR AUC Davis	0.1875	0.2443	<u>0.2159</u>
CWA	0.5397	0.6647	0.625
Kappa	0.3579	0.5284	0.5568
MCC	0.3352	0.4829	0.534
H-measure	0.1761	0.2556	0.534
Brier loss	0.3579	0.4829	0.5795
Log loss	0.5284	0.4943	0.7954
Krippendorff's Alpha	0.71801	0.62049	0.50324

3.3.2 Results

3.3.2.1 Global results Our initial results involve computing the score of each one of the 4 classifiers using all 14 evaluated metrics on the 66 selected datasets. As previously mentioned, we then transform these values into rankings and compute the MD for each performance metric. Table 5 presents these results for the Wisconsin dataset, the second least imbalanced dataset out of the 66 evaluated, which has been included for demonstration of

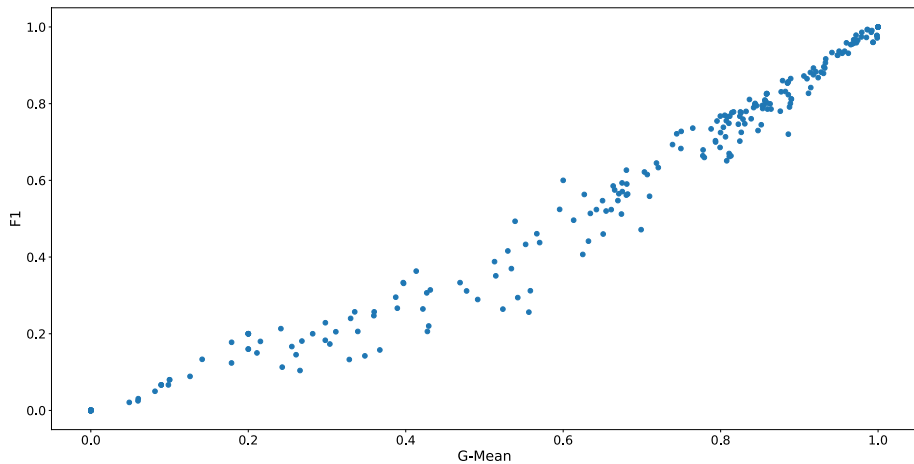


Fig. 7 Pairwise comparison between G-mean and F_1

Table 7 MD of threshold metrics averaged by imbalance class (best score by imbalance level is underlined; best overall score is in boldface)

Metric	MD average		
	$IR < 9$	$9 \leq IR < 13$	$13 \leq IR$
F_1	<u>0.0227</u>	0.0113	0.1136
G-mean	0.2954	0.284	0.2272
CWA	0.1818	0.2613	0.0909
Kappa	0.0454	0.0568	<u>0.0227</u>
MCC	0.0681	0.1704	0.0681
Krippendorff's Alpha	0.91623	0.87477	0.90759

the calculation of the MD. We do not include all the results for the remaining datasets here due to space, readability, and interpretability constraints,⁴ but we will observe later that the differences in metrics' rankings are more pronounced on datasets presenting a higher level of imbalance. As such, the results are then combined using an arithmetic average of the MD over each class imbalance ratio, as defined in Sect. 3.1: slightly ($IR < 9$), moderately ($9 \leq IR < 13$), and heavily ($IR \geq 13$) imbalanced. Table 6 presents these aggregated results for all of the performance metrics.

As per our previous work's conclusions, we can confirm the fact that the different interpolations of the same metric and other metrics associated with the PR curve (PR AUC with Linear Interpolation, Interpolated Precision AUC, AP, PR-Gain AUC, and PR AUC Davis) all present similar rankings, as their MD is averaging lower scores. This means that these metrics present a higher level of agreement with the general ranking consensus between all tested metrics. Furthermore, we remark that the G-Mean, the

⁴ Availability of data and material: The results on other datasets as well as stratified cross-validated results, that were not included in the manuscript, are available at the following link <https://github.com/jgaud/PerformanceEvaluationImbalancedClassification>.

Table 8 MD of ranking metrics averaged by imbalance class (best score by imbalance level is underlined; best overall score is in boldface)

Metric	MD average		
	IR < 9	9 <= IR < 13	13 <= IR
ROC AUC	<u>0.1193</u>	0.2727	0.2045
PR-gain AUC	0.1534	<u>0.2045</u>	<u>0.1818</u>
PR AUC Davis	0.2102	<u>0.2045</u>	0.3863
H-measure	0.1534	<u>0.2045</u>	0.2045
Krippendorff's Alpha	0.86932	0.81534	0.75710

F_1 measure, and other newly added metrics, namely the Kappa score, MCC, Brier loss, and log loss all present a higher level of disagreement, showing that these metrics provide considerably different ranking results.

Another interesting observation is that the ROC AUC and H-Measure metrics, while not being as low as their PR AUC counterparts, both present lower levels of disagreement between each other than other metrics. We believe that this behavior can be interpreted by two explanations: (i) being both rank-based metrics, they are more likely to give a similar ranking to the PR AUC interpolation methods and thus, obtain a higher level of agreement; and, (ii) as discussed in Sect. 2, the H-measure is actually based on the ROC curve for its computation.

3.3.2.2 Threshold metrics Table 7 displays the MD averages and Krippendorff's Alpha computed only on the five threshold metrics: F_1 , G-Mean, CWA, Kappa, and MCC. We notice here, as expected, a consensus of all metrics, which leads to an overall lower MD than what we previously saw in the experiment with all metrics. An interesting observation here is that while we previously mentioned that the disagreement between different metrics tends to increase in correlation with the increase in the imbalance of the dataset, this behavior does not seem to be as pronounced when using threshold metrics exclusively. Indeed, we can see with both the MD and Krippendorff's Alpha, that the agreement level stays in fact relatively constant across imbalance levels compared to the previous results and that it is at its lowest in the moderately averaged imbalance setting, mainly caused by G-Mean and CWA. Both of these metrics use specificity, which considers the true negatives of the confusion matrix in the computation of their value, while the others do not. The following logical explanation is that giving weight to the majority (negative) class using the true negative value, appears to change the rankings of the classifier in a notable way compared to other metrics that do not. This behavior could be desirable or not depending on the context of the problem, as we will discuss later on. Moreover, looking more closely at the relationship between one of these metrics, G-Mean, and the F_1 score in Fig. 7, we see that while the score of the two metrics seems to be closely related at higher scores, it does deviate much more at lower ones. Indeed, G-Mean tends to give a much higher score than F_1 on classifiers that are performing worse. Again, the fact the G-Mean also gives equal importance to the majority class by using the true negatives would explain this behavior.

3.3.2.3 Ranking metrics For ranking metrics, being that we analyzed multiple implementations of the same performance metric (PR AUC) and we expect these implementations to have similar results as per our previous work (Gaudreault et al., 2021), we chose only one metric to represent all of the PR AUC implementations, namely Davis' interpolation. This implemen-

Table 9 Agreement of probabilistic metrics averaged by imbalance class

Metric	IR < 9	9 ≤ IR < 13	13 ≤ IR
Krippendorff's Alpha	0.86477	0.85681	0.75341

Table 10 MD of all PR AUC implementations averaged by imbalance class (best score by imbalance level is underlined; best overall score is in boldface)

Metric	MD average		
	IR < 9	9 ≤ IR < 13	13 ≤ IR
PR AUC (linear interpolation)	0.0909	0.034	0.125
Interpolated precision AUC	0.1477	0.1477	<u>0.1022</u>
AP	0.1818	0.1022	0.3522
PR AUC Davis	<u>0.0795</u>	0.0113	<u>0.1022</u>
Krippendorff's Alpha	0.90767	0.94602	0.86506

tation was chosen due to it being the most consistent with the consensus of the group in our previous work's experiments, a conclusion which we will also verify in the next section.

Table 8 presents the MD Average and Krippendorff's Alpha over these metrics. In this case, in contrast with the threshold metrics previously tested, there is a more noticeable decrease in agreement between the metrics with the increase in imbalance, as seen by the Krippendorff's Alpha. Interestingly, there does not seem to be any metric that stands out of the others with its level of agreement. For example, the ROC AUC does display the most agreement with the common consensus in the lowest imbalance setting, but not in the intermediate one and highest one where PR-Gain AUC is relatively low. It is likely that these ranking metrics evaluate a model quite differently depending on its context and dataset imbalance. For example, this behavior was previously demonstrated by Hand (2009) between the H-Measure and ROC AUC and by Davis and Goadrich (2006) between the ROC and PR AUCs.

3.3.2.4 Probabilistic metrics Since we only included two different probabilistic metrics, computing the MD for these would not make sense as they will both get the same value. Nonetheless, we can still compute the Krippendorff's Alpha which measures the general agreement between them at each imbalance level, the results of which are displayed in Table 9. In this case, we observe once again a decrease in agreement as the imbalance between the classes' distribution increases.

3.3.2.5 PR AUC implementations In this section, we test the agreement of the different calculations of the PR AUC in order to determine if the use of one implementation over another would lead to different results in the perceived performance of a model compared to another.

The results in Table 10 show that across all levels of imbalance, the agreement of all metrics stays relatively high and only drops at higher imbalance ratios, mainly due to AP. These results are in line with our previous conclusions (Gaudreault et al., 2021), as it shows that the different interpolations of the PR AUC do lead to very similar results compared to the difference between entirely different metrics for example. Moreover, PR AUC Davis exhibits the highest level of agreement across all imbalance ratios which is why we chose it as our baseline to represent the PR AUC metric in other experiments and why we recommend it as a good implementation to compute the PR AUC as it depicts a good overall representation.

Table 11 MD of threshold and ranking metrics averaged by imbalance class (best score by imbalance level is underlined; best overall score is in boldface)

Metric	MD average		
	IR < 9	9 ≤ IR < 13	13 ≤ IR
F_1	0.1875	<u>0.1647</u>	0.1988
G-mean	0.4147	0.4147	0.2897
ROC AUC	0.4943	0.5397	0.9147
PR-gain AUC	0.4943	0.6079	0.8238
PR AUC Davis	0.3806	0.6079	0.6306
CWA	0.3011	0.3238	0.1988
Kappa	0.1193	<u>0.1647</u>	0.1193
MCC	<u>0.1079</u>	0.1875	0.0965
H-score	0.4034	0.5397	0.9488
Krippendorff's Alpha	0.72699	0.63443	0.53230

3.3.2.6 Threshold and ranking metrics Finally, we inspect in Table 11 both threshold and ranking metrics together, choosing again PR AUC Davis in order to represent the different PR AUC implementations for the same reasons stated above.

First of all, we can note the much higher agreement of threshold metrics with the consensus compared to ranking metrics. As we saw before, when analyzing each group individually, the agreement of threshold metrics was higher between themselves than ranking ones, especially at higher imbalance ratios. Moreover, these results show a significant decrease in the agreement between all metrics as the imbalance increases. In the highest imbalance group, most ranking metrics disagree the most with the consensus while threshold ones agree the most. This observation solidifies the conclusion and recommendation that when dealing with highly imbalanced data, the use of multiple performance metrics, particularly ones that are fundamentally different is strongly recommended, as they exhibit higher variance in their results.

Secondly, we can observe again the cluster formed between F_1 , Kappa, and MCC presented in Sect. 3.2 when using hierarchical clustering. Indeed, these three metrics seem closely related, as we observe a fairly high level of agreement at all imbalance levels.

3.4 Summary

This section summarizes the multiple conclusions reached in Sect. 3, as we evaluated the impact of using different performance metrics on the ranking of different models.

- The three different types of metrics (threshold, ranking, and probabilistic) exhibit very different behavior, and are generally highly correlated with other metrics of their own group.
- Threshold and ranking metrics are much more closely related between themselves than either of them are to probabilistic metrics. However, this correlation does decrease when the imbalance of the data increases, which shows the importance of using multiple types of metrics, especially in high imbalance contexts.
- When looking at threshold metrics specifically, MCC and Kappa give extremely similar results, but Kappa does exhibit some unwanted properties when dealing with imbalanced domains which means that the use of MCC should be preferred.

- The different implementations of the PR AUC (Linear Interpolation, Interpolated Precision, AP, and Davis) are very close to one another, with AP exhibiting more disagreement than the others, while PR AUC Davis comes out on top in relation to it consistently agreeing with the other implementations, which is why it should be favored when computing the PR AUC.
- PR-Gain displays different properties than the different computations of the PR AUC and should therefore be treated as a different metric.

4 Evaluation of the impact of noise

In this section, we study and analyze how the different performance metrics we selected are affected by different intrinsic characteristics of the data, such as noise and class imbalance. Depending on the data and context of deployment, these characteristics can be present and can potentially affect the ability of certain metrics to accurately give insights into a model's performance. As such, we want to provide future insights and recommendations for researchers and computer scientists to choose the right metric for their application. This section will be divided into two subsections described as follows: (i) presentation of the methodology used throughout these experiments; and (ii) analysis and discussion of the results.

4.1 Methodology

The problem of evaluating the impact of noise and other dataset properties on performance metrics is complex, as introducing noise and modifying the dataset directly can impact a machine learning model as well as the performance metric which would render the results unusable for our purpose. For example, a given classifier could perform worse in a scenario where noise is introduced in the data and there would be no way to determine the impact of the noise on the metric itself rather than the classifier. Therefore, we needed to devise a way to isolate the impact on the performance metrics alone. To do so, we adopted an approach similar to the one presented by Ferri et al. (2009), which will be presented in the remainder of this section.

First, we start by creating a synthetic dataset containing 1000 samples with the specific characteristics we want to test in a given experiment. Then, a classifier C_1 is artificially created on this data, by simply generating probabilities for each y sample, using random values in a uniform distribution between $[0, 0.5]$ or $(0.5, 1]$, depending on the true value of y . A total of 10% of these probabilities (100 samples) are then reshuffled, now using a random value within the uniform distribution of $[0,1]$. In other words, we are artificially creating a classifier that is very accurately predicting the data created, this is done so in order to eliminate the impact that the classifier has and focus solely on the impact on the performance metrics since we are not interested in evaluating the classifier's robustness to noise, but rather the metric's. Finally, we define a second classifier C_2 , by using the same probabilities as C_1 , but changing 10% of all probabilities (100 samples) to strictly the opposite of the y true value, essentially creating a strictly worse classifier. This means that at the start of the experimentation, the classifier C_1 will always be better performing than the classifier C_2 .

For each experiment, we perform three different tests, creating the data with an IR of 4.5, 11, and 40, which corresponds to a value near the median of each imbalance ratio

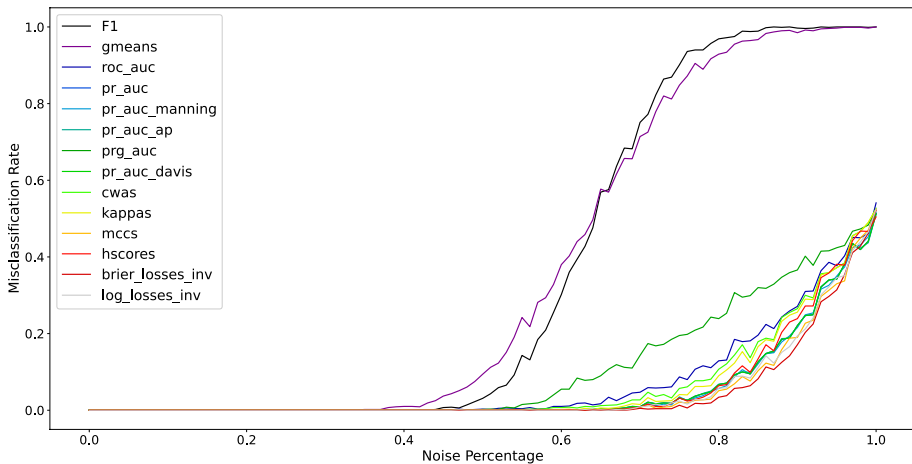


Fig. 8 Performance metrics robustness to misclassification noise on slightly imbalanced data (IR = 4.5)

interval previously defined in Sect. 3 (slightly, moderately, and heavily imbalanced). The noise level is changed gradually throughout these tests and we observe which classifier (C_1 or C_2) achieves a better score for each performance metric. A score of 1 is added if the metric selects the classifier C_2 rather than C_1 , a score of 0 for the opposite, and a score of 0.5 in case both are equal. Each of those tests is performed for a total of 1000 trials, where a new dataset and classifiers are created each time. All trials are then averaged together, which ensures the convergence of the results.

We performed several experiments in order to test the impact of misclassification and probability noises on the performance metrics. As defined in Ferri et al. (2009), we used two types of noise to be tested, as described below, however, our experiments differ in several ways, as we focus specifically on performance metrics applicable in imbalanced scenarios and consider the imbalance ratio of the datasets for each experiment.

- *Misclassification Noise*: By applying noise to the true labels by either changing its value to its opposite or randomly, we can evaluate the sensitivity of the metrics to changes in the true labels.
- *Probability Noise*: By applying noise directly to the probabilities of the classifier, we measure the sensitivity of the metric to changes in model probabilities.

4.2 Results

4.2.1 Misclassification noise

Misclassification noise is defined as noise affecting the data classes directly. In practice, this could be represented by noise in the deployment context of a model or labeling errors for example. Using a random value from a normal distribution, we affect the y true values of our testing scenario we have presented in the previous section, from a range of 0%, where no y values are changed, to 100%, where all the values would be changed randomly. Consequently, we expect that there would no longer be a distinction of performance

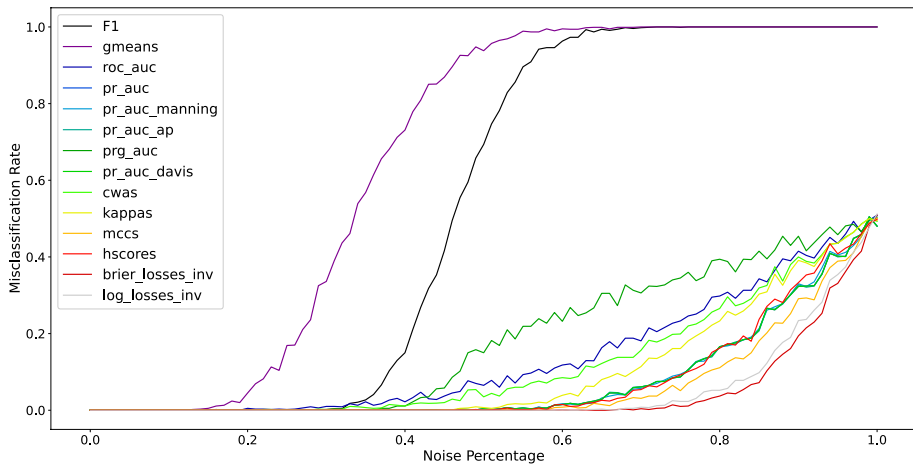


Fig. 9 Performance metrics robustness to misclassification noise on moderately imbalanced data ($IR = 11$)

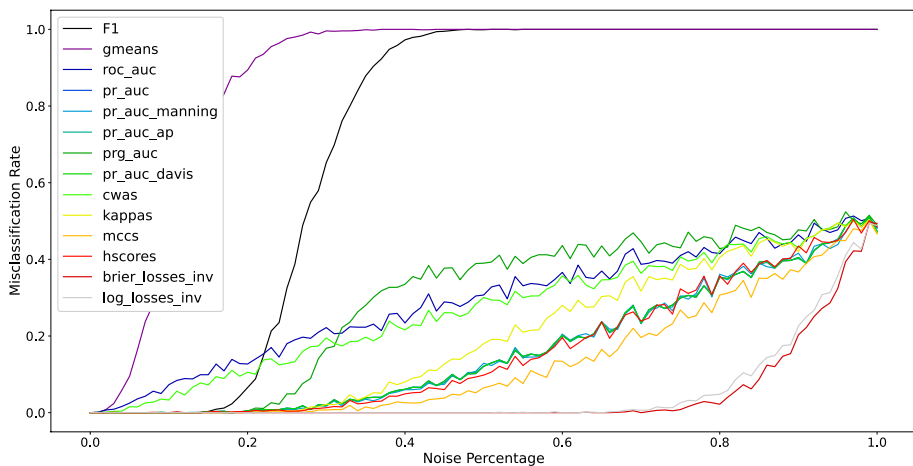


Fig. 10 Performance metrics robustness to misclassification noise on heavily imbalanced data ($IR = 40$)

between the two classifiers, C_1 and C_2 , as the noise reaches 100% since all labels will now be randomly chosen.

Figures 8, 9, and 10 display the results for each imbalance group, slightly ($IR = 4.5$), moderately ($IR = 11$), and heavily imbalanced ($IR = 40$), respectively. While the x-axis represents the percentage of noise applied to the data as previously mentioned, the y-axis represents the percentage of times classifier C_2 was "misclassified" as the better classifier rather than classifier C_1 by a given metric, which we will refer to as the *misclassification rate* from now on.

As we expected, all metrics start with a misclassification rate of 0, as there is no noise added to the data yet. As more and more noise is added, we expect the metrics to trend towards a rate of 0.5 since there would be no real distinction between the two classifiers considering all the class labels are getting randomized. In that regard, most metrics do

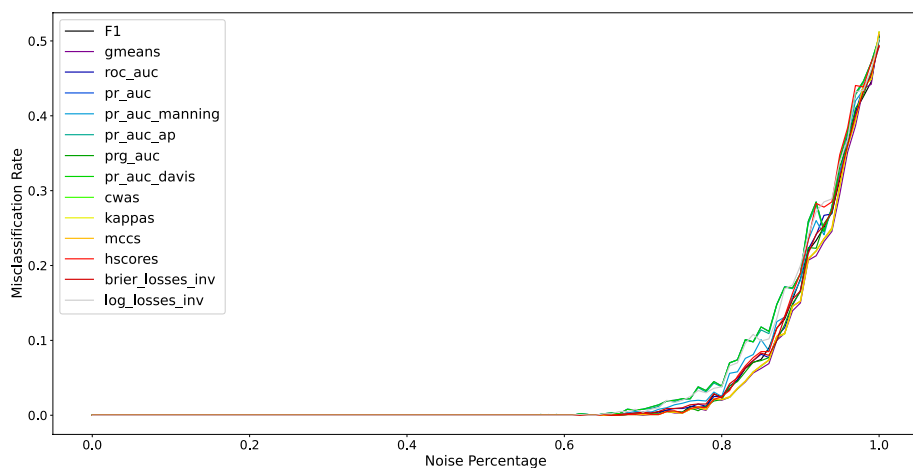


Fig. 11 Performance metrics robustness to misclassification noise on balanced data (IR = 1)

follow that hypothesis, for all the imbalance groups. However, two metrics hold our attention as they do not follow this same pattern. Indeed, F_1 score and G-Mean present a distinct behavior, as both of them not only misclassify the classifiers at a lower noise amount but also trend towards classifying classifier C_2 as the best one when the dataset is comprised only of random y values. This observation signifies that both of these metrics do represent a lesser resilience to misclassification noise in general. Moreover, we also note the misclassification rate of PR Gain, which also represents a slightly higher sensitivity to this type of noise than its counterparts, albeit smaller than F_1 score and G-Mean.

Now focusing our attention on class imbalance, we note the heavy effect that it has on the misclassification rate. Undoubtedly, the increase in class imbalance causes most metrics to be less robust to misclassification noise, which make sense since any changes affecting the small number of samples of the minority class would be seen in the overall performance of the model. An interesting observation, however, is that both Brier loss and Log loss seem to be largely unaffected by the class imbalance. This can be explained by the fact that both metrics are using the predicted probabilities of the model rather than the classification prediction errors which would not be as affected by further imbalance in the data.

Finally, we also wanted to note the disparity of our results with the experiments performed by Ferri et al. (2009). Indeed, in their case, all metrics performed much more similarly, reaching an average of 50% misclassification rate at 100% noise. To validate our results, we carried out an experiment similar to our previous one but with all datasets balanced (IR = 1). When performing our experiments on balanced data, we are able to confirm the results obtained in Ferri et al. (2009) as we observe similar trends which can be seen in Fig. 11. However, we can observe that in our results, the misclassification rate only starts increasing at around 80% noise which is much later than the results shown in Ferri et al. (2009). This is explained by the fact that we made our classifier C_2 strictly worse than C_1 at the beginning, by changing the predictions to their opposite rather than randomly like it was done in the aforementioned study, which therefore needs a higher percentage of random noise in order for C_2 to outperform C_1 . Compared to the balanced scenario, the imbalance cases showed that all metrics are actually much more sensitive to misclassification noise when data imbalance is present and that metrics such as F_1 and G-Mean should

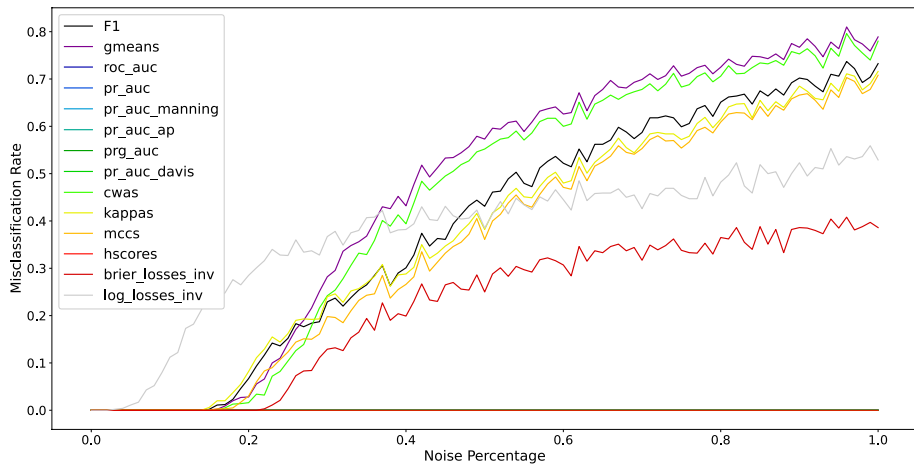


Fig. 12 Performance metrics robustness to probability noise with IR = 4.5

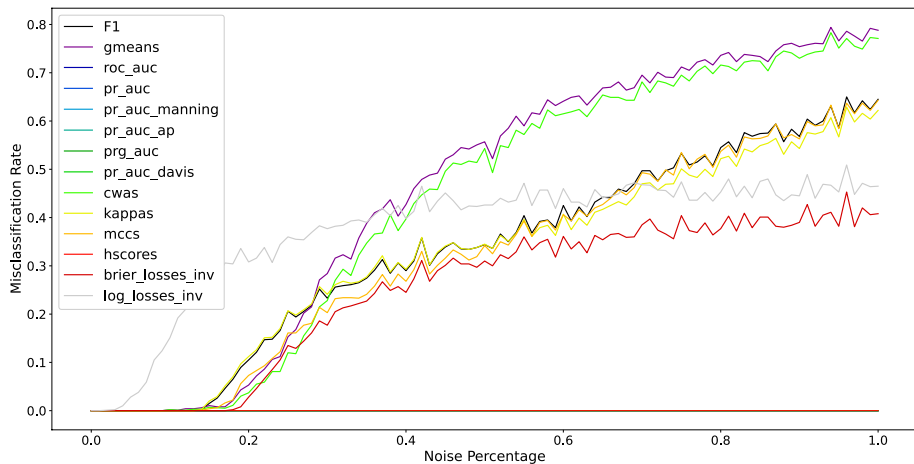


Fig. 13 Performance metrics robustness to probability noise with IR = 11

be avoided in these scenarios, as they present a much higher sensitivity even at low imbalance levels. These results show the high impact that class imbalance has on all performance metrics and highlights the importance of performing these experiments specifically on imbalanced data to provide insights for users in this domain.

4.2.2 Probability noise

In this section, rather than evaluating noise on the class values, we are evaluating the impact of noise on the probabilities from the classifiers directly. In other words, these experiments test the resilience of the performance metrics to poor prediction probabilities given by the classifier. To test the impact of this type of noise on our performance

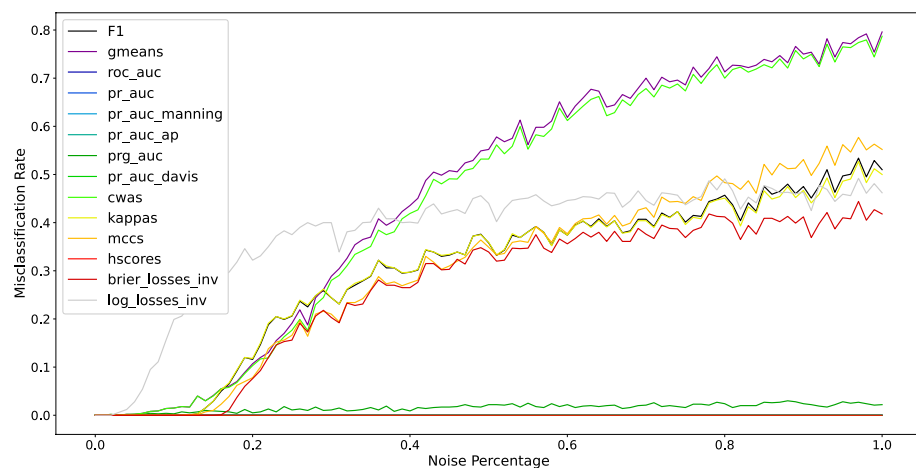


Fig. 14 Performance metrics robustness to probability noise with IR = 40

metrics, we first start by generating a value α , from a uniform distribution ranging from $[-\beta, \beta]$ where the value of β depends on the amount of noise being added and ranges from 0 to 0.5, depending on the percentage of noise, where a value of 0 represents 0% of noise and 0.5, 100%. The value of α will then be added uniformly to all the probabilities of classifier C_1 , while the probabilities of classifier C_2 will remain the same. In cases where the value of a given changed probability would reach a value outside of the range $[0, 1]$, the value is pinned to the corresponding minimum or maximum instead.

The results of this set of experiments can be observed in Figs. 12, 13, and 14 for the slightly, moderately and highly imbalanced data, respectively. In all cases, we observe that there are multiple metrics that seem to be unaffected by the introduction of probability noise. Indeed, we note that all the ranking metrics tested have largely not been influenced by the probability noise, while threshold and probabilistic metrics such as G-Mean, CWA, log loss, and others are significantly affected. This behavior can be explained by the fact that affecting probabilities in such a way that all probabilities are changed by a common value, does not generally affect their rankings, which is why ranking metrics are largely unaffected. On the other hand, since threshold metrics are based on the classification prediction errors and probabilistic metrics on the probabilities themselves, both types of metrics are heavily affected by the probability noise which ultimately makes the difference, or boundary between both predicted classes harder to distinguish.

Looking at the difference between imbalance groups, we can see similar behavior to what we observed previously with misclassification noise in Figs. 8, 9, and 10, where most metrics are more sensitive to the noise on higher imbalance ratios. Indeed, this effect is also present here, albeit to a lesser degree, as we can see that the metrics that were affected by the noise in lower imbalance scenarios start to be affected at lower levels of noise in the higher imbalance cases which restates one of the numerous difficulties when dealing with imbalanced data.

4.3 Summary

This section summarizes the multiple conclusions reached in Sect. 4, as we evaluated the impact of misclassification and probability noise on the metrics at different imbalance levels.

- F_1 score and G-Mean showed the least resilience to misclassification noise on imbalanced data. While lesser than the two aforementioned metrics, PR-Gain AUC also performed worse with the introduction of misclassification noise than the other evaluated ranking metrics.
- The metrics' sensitivity to misclassification noise was demonstrated to be largely driven by class imbalance, as it not only increased with class imbalance but was also much less present on balanced data.
- Ranking metrics proved to be largely unaffected by noise on the model's probabilities (probability noise), while threshold and probabilistic metrics were to a much greater extent.
- While class imbalance did not play as much of a role in the sensitivity of the metrics to probability noise compared to misclassification noise, it did increase it slightly on metrics that already showed sensitivity to this type of noise in other scenarios with lesser imbalance.

5 Conclusions and recommendations

Throughout this study, we analyzed multiple performance metrics widely regarded as the given metrics to use when dealing with imbalanced domains. Using the knowledge acquired through our experiments and the conclusions reached, we devised the data flow diagram shown in Fig. 15 with the goal of helping future researchers, computer scientists, and end-users to choose an appropriate metric depending on the specific context of their applications and use cases. It is important to note that while the following diagram suggests one or a few metrics for each specific scenario, these are merely intended as a starting point and should therefore be treated as such. Moreover, we highly recommend the use of multiple metrics, notably ones that come from different types (threshold, ranking, and probabilistic) to validate the evaluation of a model, since they can present very different behavior, as we have seen in Sect. 3.

The recommendations given in the diagram follow the conclusions we have previously reached and discussed which can be summarized as follows:

1. General Conclusions

- (a) The three different types of metrics (threshold, ranking, and probabilistic) exhibit very different behavior and their usage should be complementary, especially when dealing with highly imbalanced data.
- (b) Threshold and ranking metrics are much more closely related to each other than either of them are to probabilistic metrics. This indicates that probabilistic metrics are a class apart from the two other ones, which is why we only recommended them for specific scenarios where the probabilities of the model are important in the diagram.

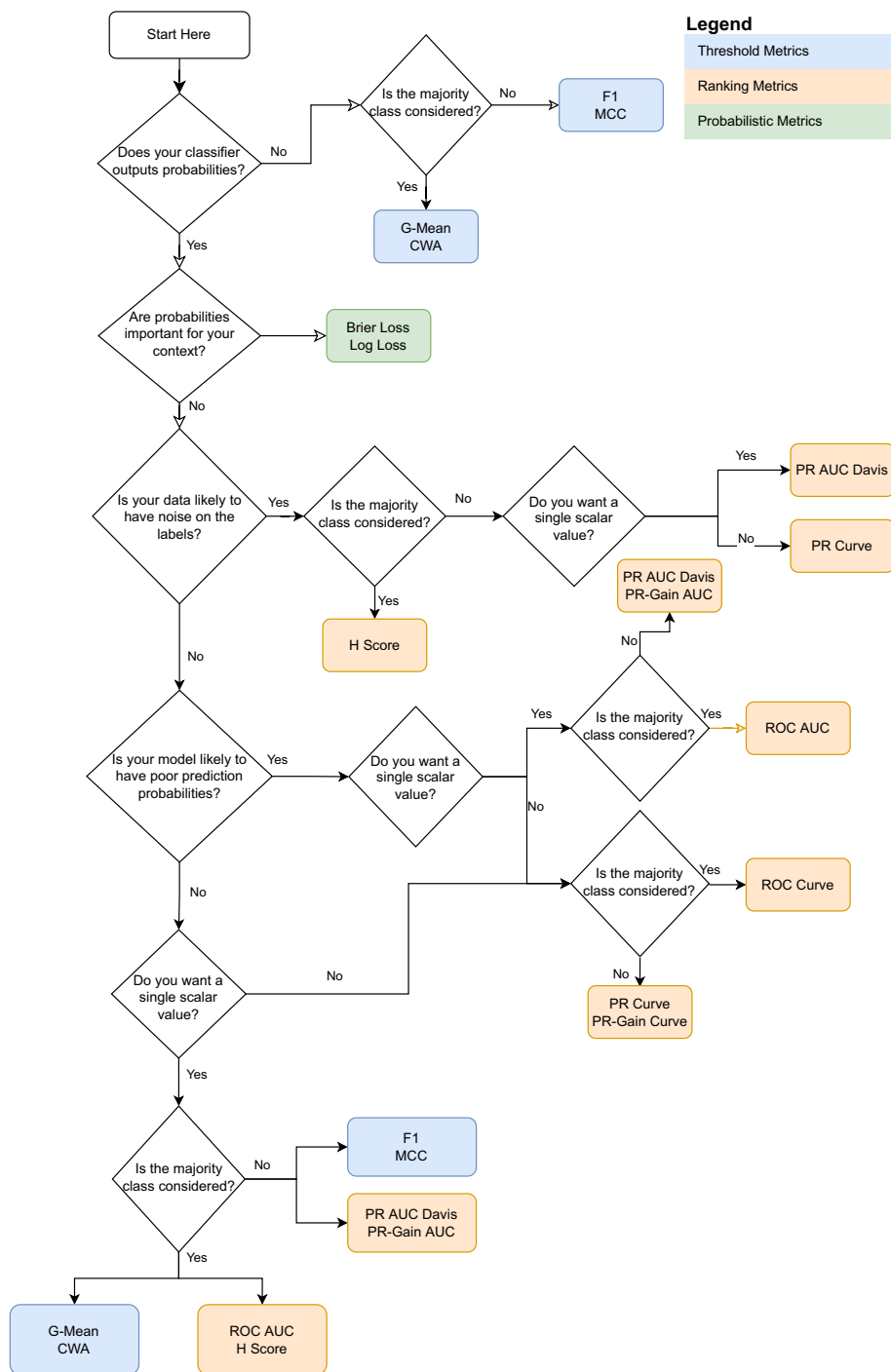


Fig. 15 Data flow diagram representing recommendations for performance metrics choice in imbalanced domains

- (c) In situations where MCC and Kappa are applicable, we have recommended the use of MCC over Kappa since our results presented that both metrics were mostly similar but, as we have discussed, Kappa does display some unwanted properties when it is used in imbalanced domains.
- (d) Davis' interpolation of the PR AUC is recommended when a scalar value representing the PR curve is wanted, as we have shown that it was the most consistent with the other interpolations of the PR AUC.
- (e) In situations where both the PR curve, PR-Gain curve, or their AUCs are applicable, we recommended the use of both or either, since we have shown that they present different characteristics and can lead to different ranking results.

2. Conclusions Related to Noise

- (a) F_1 , G-Mean, and PR-Gain are not recommended when the data is likely to have noise on the labels (misclassification noise) since it was demonstrated that they display a lesser resilience to this type of noise.
- (b) We showed that most metrics present a higher sensitivity to misclassification noise on a higher imbalance of the data. We would especially recommend avoiding the aforementioned F_1 , G-Mean, and PR-Gain in these scenarios. Moreover, we would strongly recommend combining multiple other metrics to ensure a reliable performance evaluation.
- (c) When dealing with models likely to have poor prediction probabilities (probability noise), we do not recommend the use of threshold or probabilistic metrics, as they displayed a higher sensitivity to this type of noise at all imbalance levels. We instead recommended ranking metrics, or their AUC when a scalar metric is needed.

The work in this paper represents an important contribution to experimentally and systematically reviewing performance metrics that are appropriate for imbalanced domain classification. We studied the impact of choosing one metric over another for the evaluation of a model, as well as the resilience of these different metrics to noise, and provided future researchers with easy-to-follow recommendations that can be used when deciding the performance metric to use for their next project. This serves as a strong first step towards reaching the goal of correctly evaluating models in imbalanced binary classification. Interesting future research directions include the extension of the presented analysis to imbalanced multi-class and data stream classification. Another relevant direction is the evaluation of the impact of using post and pre-processing techniques on the evaluation of the models and, ultimately, the choice of performance metric.

Appendix 1

- *Decision Tree*: Gini criterion; Best split at each node; 2 minimum samples to split; 1 minimum sample to be a leaf node;
- *Support Vector Machine*: Radial Basis Function (RBF) kernel; Regularization parameter of 1.0;
- *Random Forest*: Gini criterion; 100 total estimators; 2 minimum samples to split; 1 minimum sample to be a leaf node;
- *XGBoost*: 100 boosting rounds;

Appendix 2

See Figs. 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 and 26.

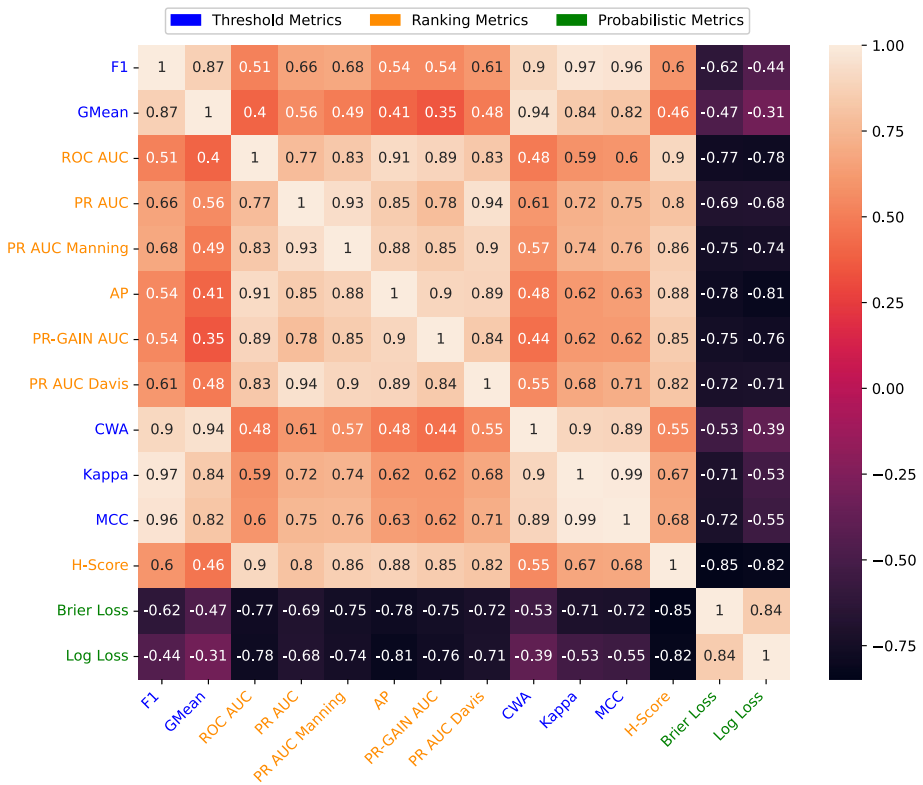


Fig. 16 Spearman correlation coefficients of all evaluated metrics over datasets with $IR < 9$

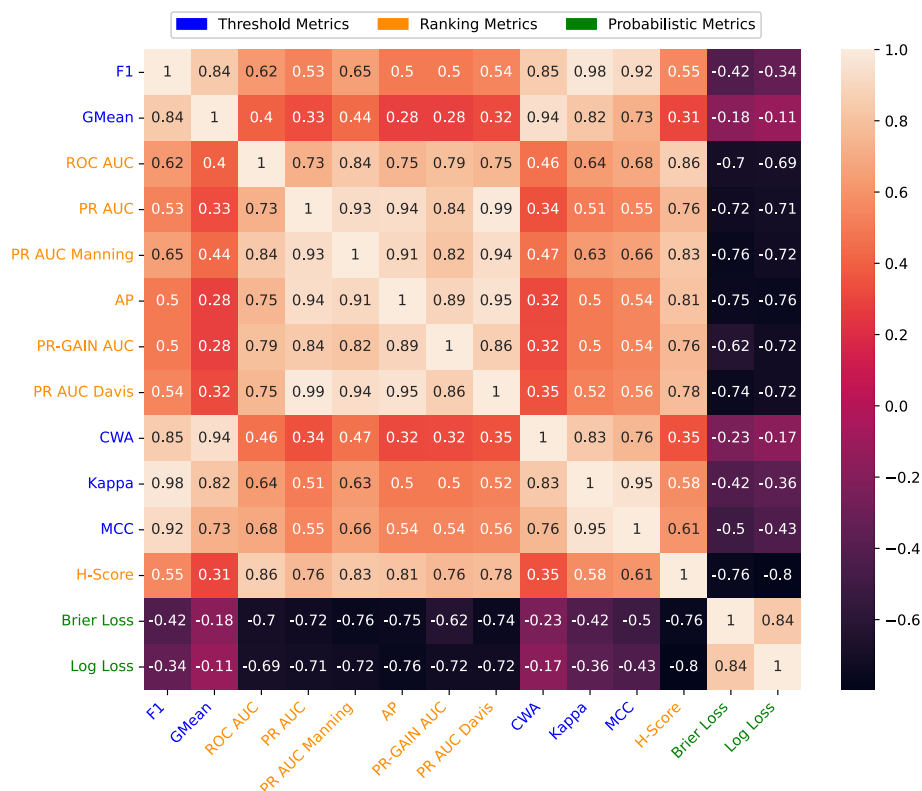


Fig. 17 Spearman correlation coefficients of all evaluated metrics over datasets with $9 \leq IR < 13$

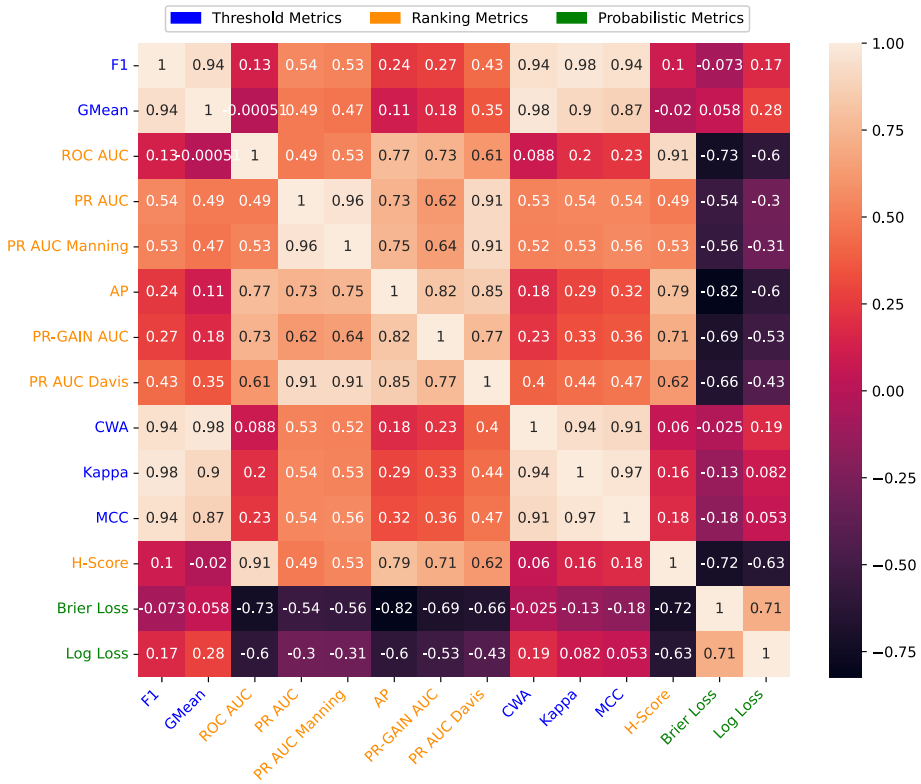


Fig. 18 Spearman correlation coefficients of all evaluated metrics over datasets with $IR \geq 13$

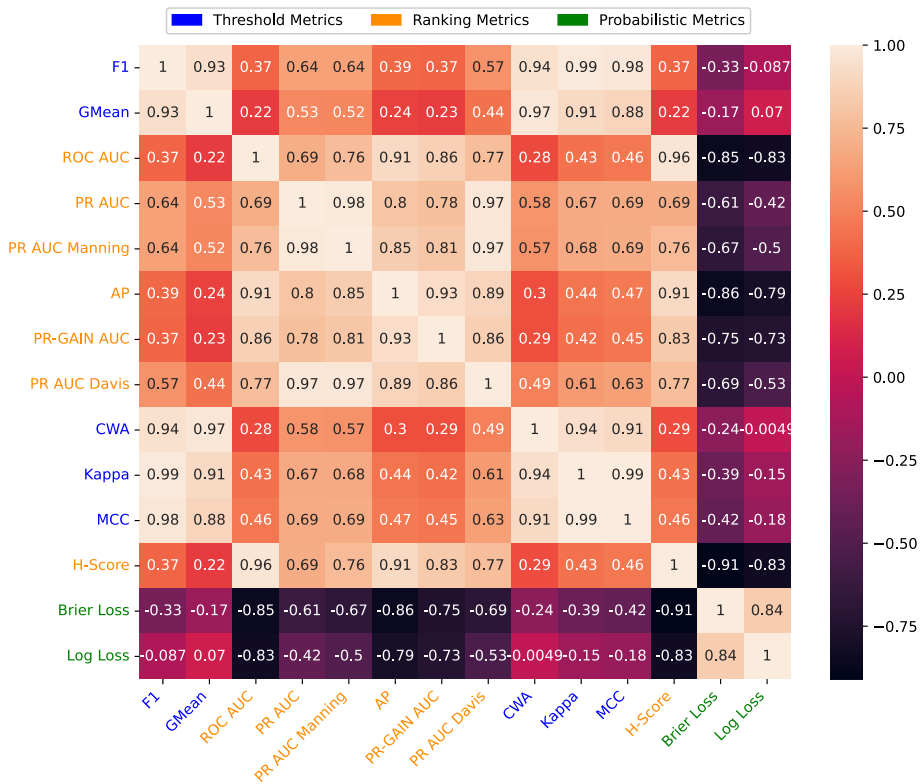


Fig. 19 Pearson correlation coefficients of all evaluated metrics over all datasets

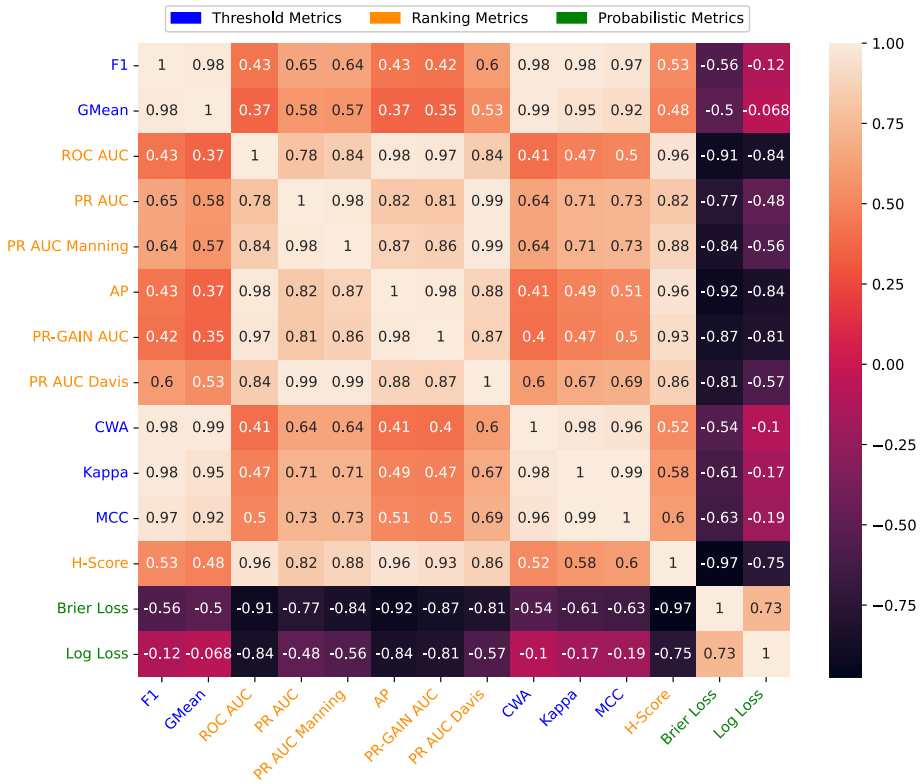


Fig. 20 Pearson correlation coefficients of all evaluated metrics over datasets with $IR < 9$

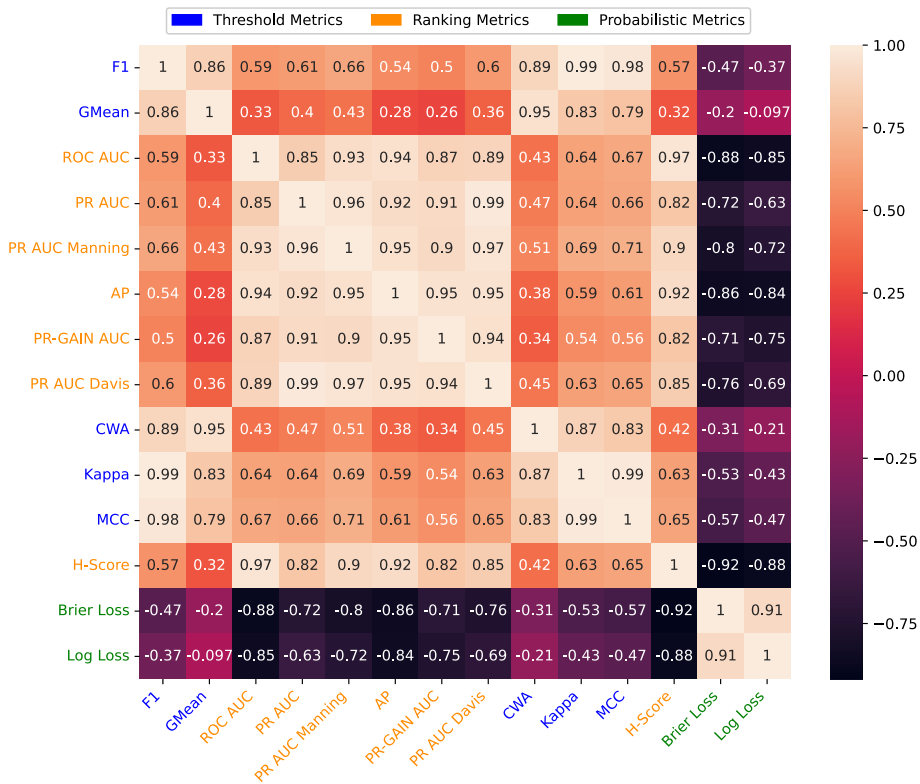


Fig. 21 Pearson correlation coefficients of all evaluated metrics over datasets with $9 \leq IR < 13$

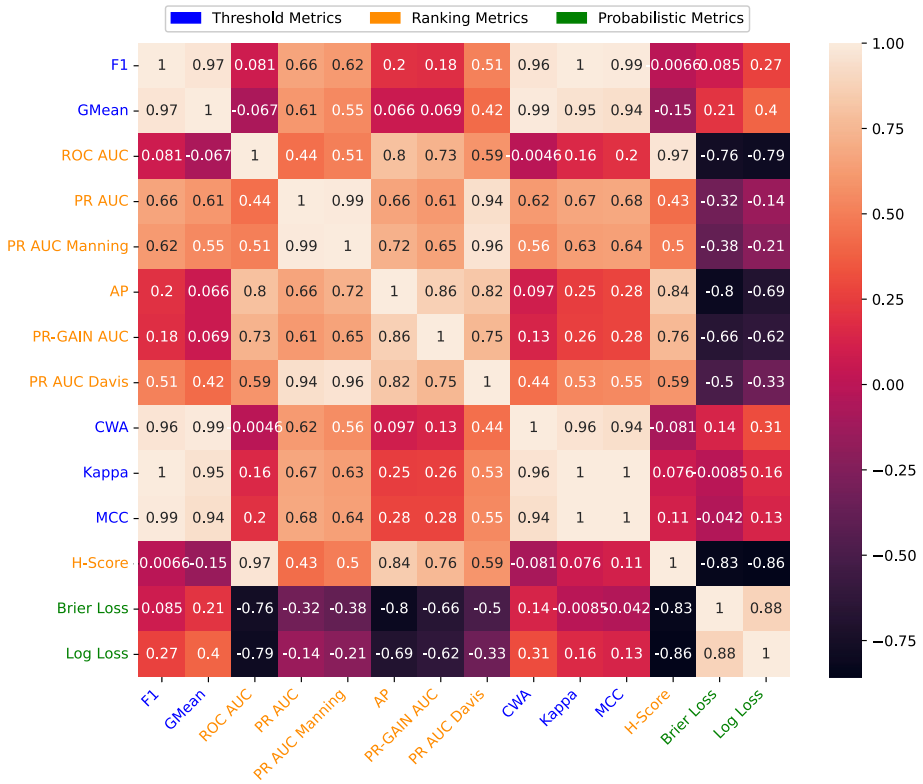


Fig. 22 Pearson correlation coefficients of all evaluated metrics over datasets with $IR \geq 13$

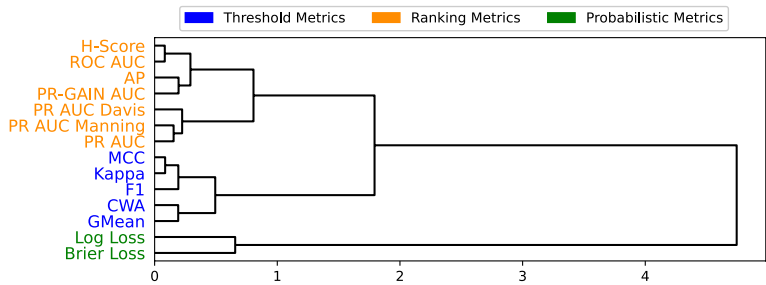


Fig. 23 Pearson hierarchical clustering of all evaluated metrics over all datasets

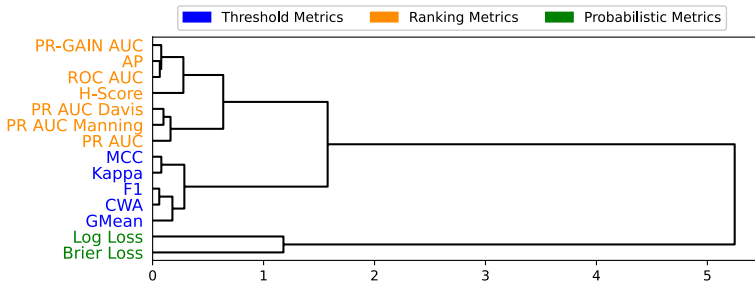


Fig. 24 Pearson hierarchical clustering of all evaluated metrics over datasets with $IR < 9$

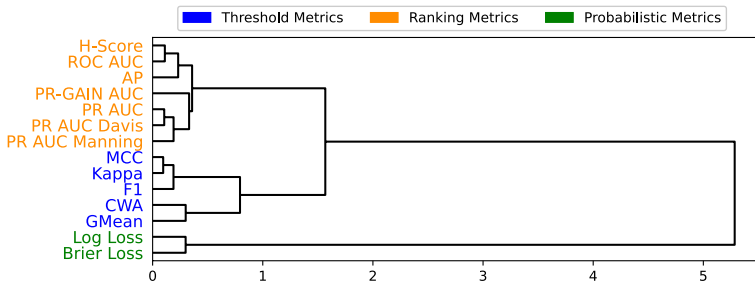


Fig. 25 Pearson hierarchical clustering of all evaluated metrics over datasets with $9 \leq IR < 13$

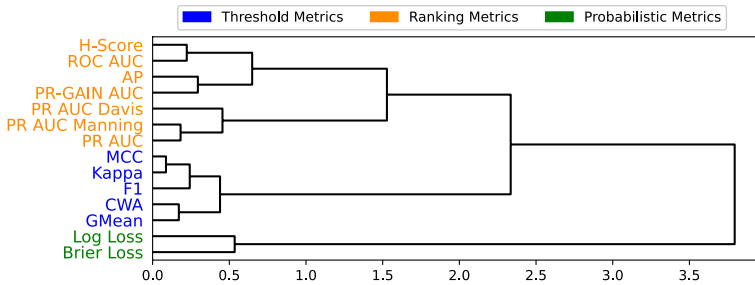


Fig. 26 Pearson hierarchical clustering of all evaluated metrics over datasets with $IR \geq 13$

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Data availability All experiments and data are publicly available (cf. <https://github.com/jgaud/PerformanceEvaluationImbalancedClassification>);

Code availability Custom code for all experiments is publicly available (cf. <https://github.com/jgaud/PerformanceEvaluationImbalancedClassification>).

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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